

Introduction to Cryptography

Dr Appala Naidu Tentu

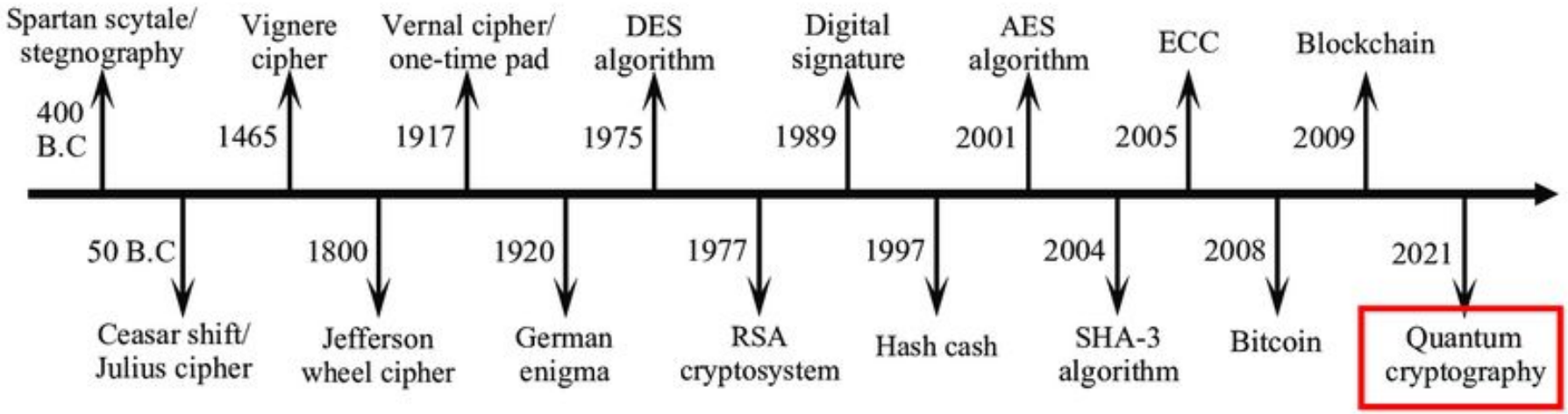
CR Rao Advanced Institute of Mathematics, Statistics and Computer Science
University of Hyderabad Campus,
Prof. C.R.Rao Road, Gachibowli, Hyderabad - 500046

Approaches to Secure Communication

- Steganography
  - "covered writing"
  - hides the existence of a message
  - depends on secrecy of method

- Cryptography
  - "hidden writing"
  - hide the meaning of a message
  - depends on secrecy of a short key, not method

Cryptography : History - Timelines



Classical Cryptography

The Problem



Bob

Private Message



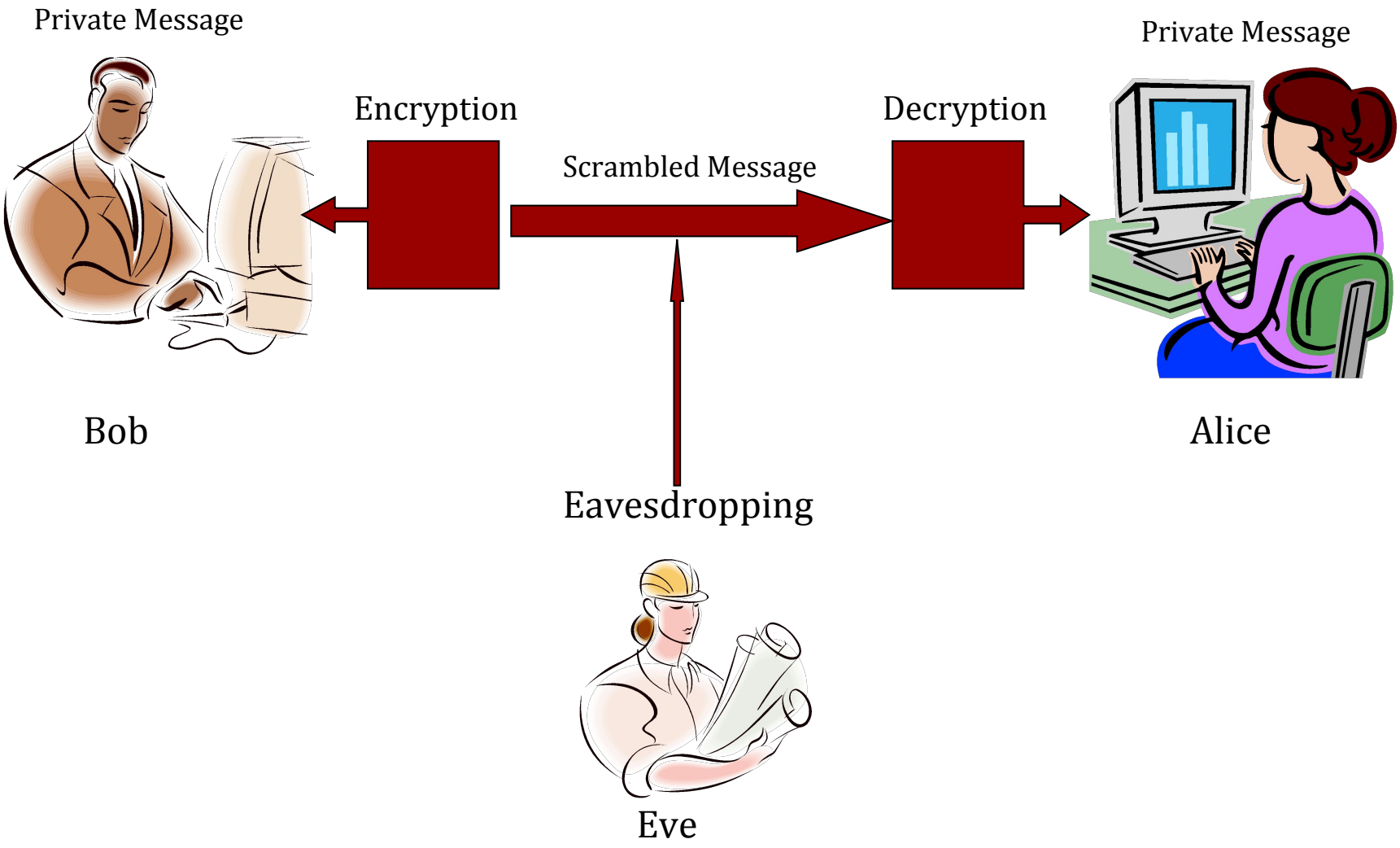
Alice

Eavesdropping



Eve

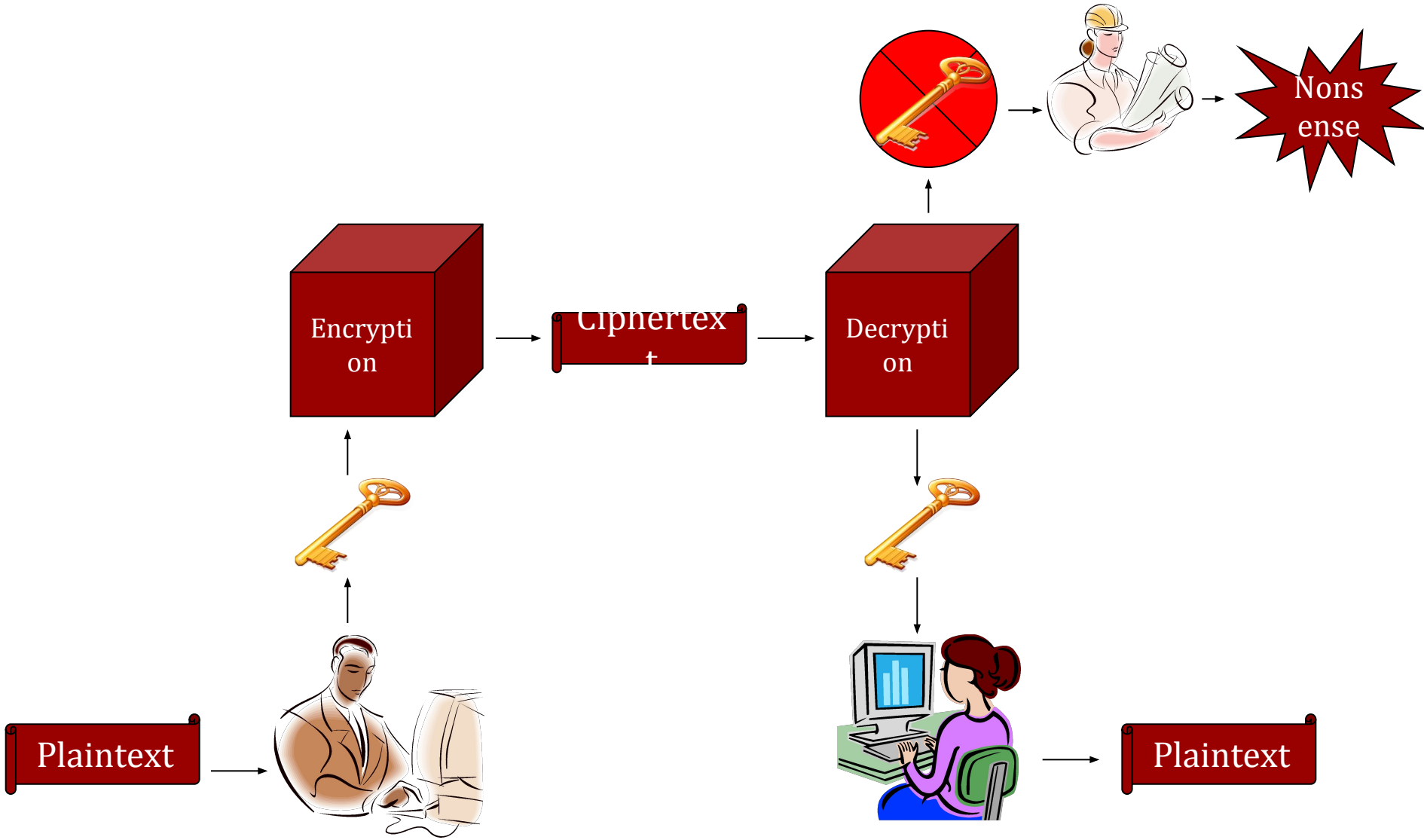
The Solution



What do we need?

- Bob and Alice want to be able to encrypt/decrypt easily
- But no one else should be able to decrypt
- How do we do this?
  - Keys!

Using Keys


Nons
ense
Encrypti
on
Ciphertex
t
Decrypti
on
Plaintext
Plaintext

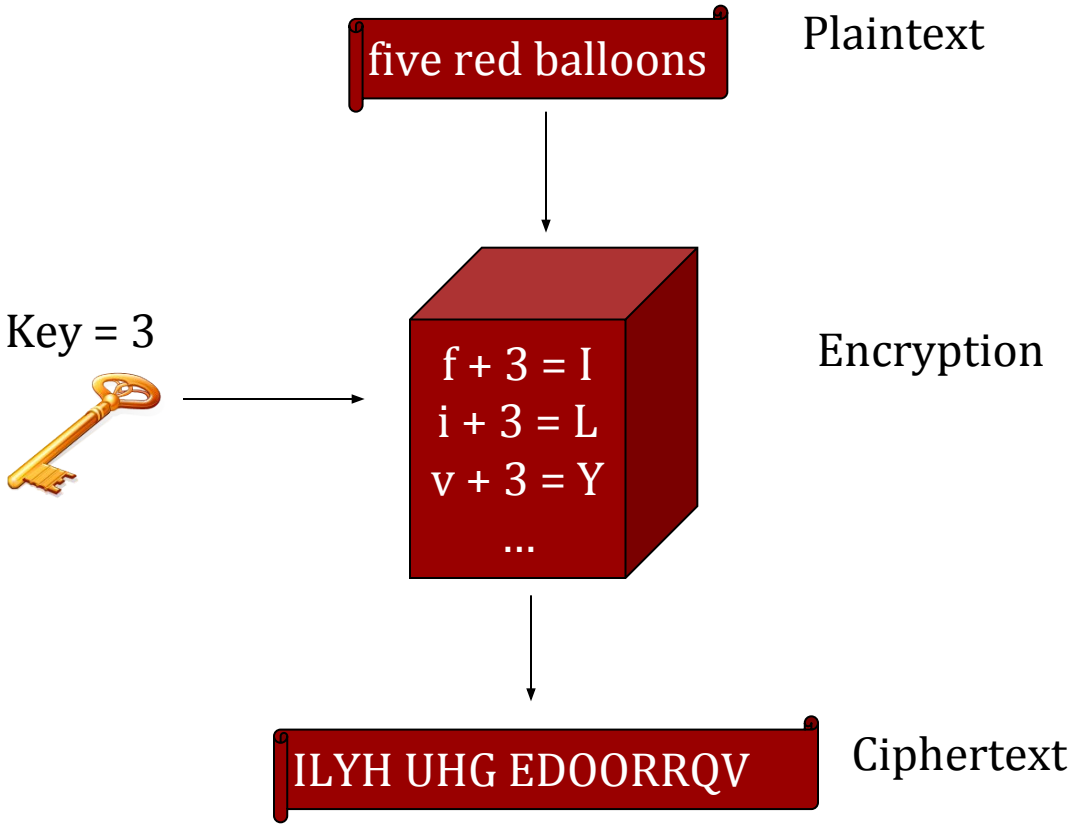
What is a cipher?

- A cipher is an algorithm for encryption/ decryption

- Kerckhoff's Principle: All details of the cipher should be public
  - Eve still shouldn't be able to decrypt messages unless she has the secret key

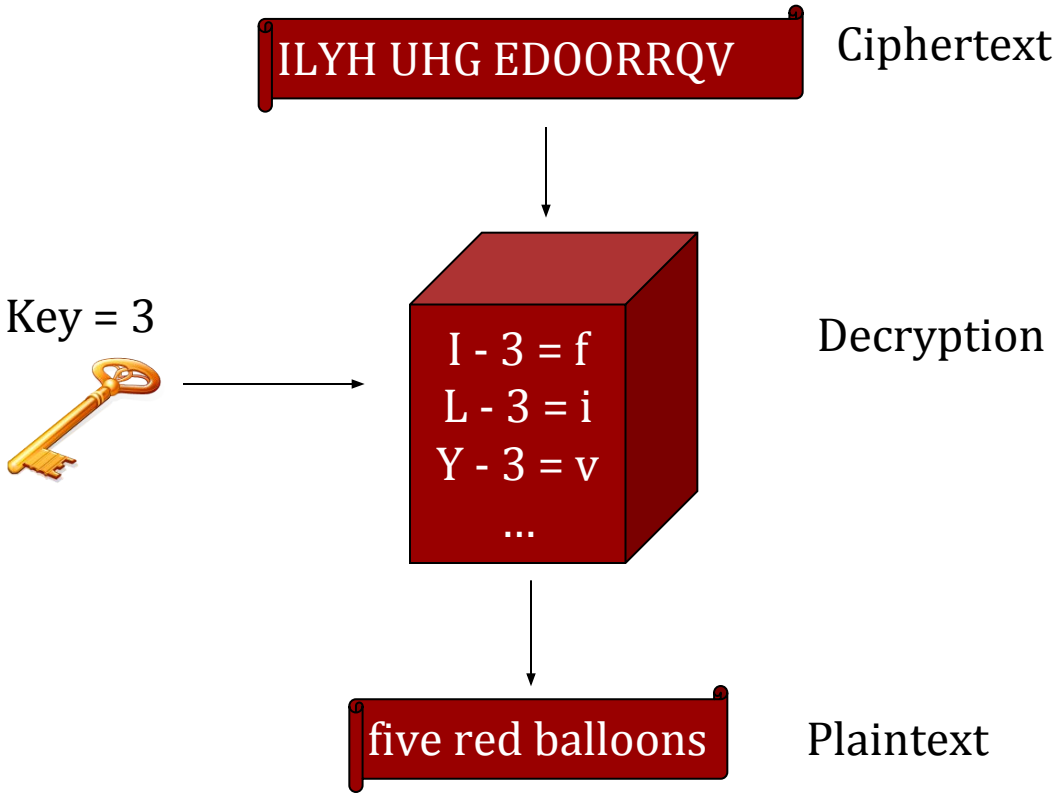
The Shift Cipher

- We “shift” each letter over by a certain amount



The Shift Cipher cont.

- To decrypt, we just subtract the key



What's wrong with the shift cipher?

- Not enough keys!
- If we shift a letter 26 times, we get the same letter back
  - A shift of 27 is the same as a shift of 1, etc.
  - So we only have 25 keys (1 to 25)
- Eve just tries every key until she finds the right one

The Substitution Cipher

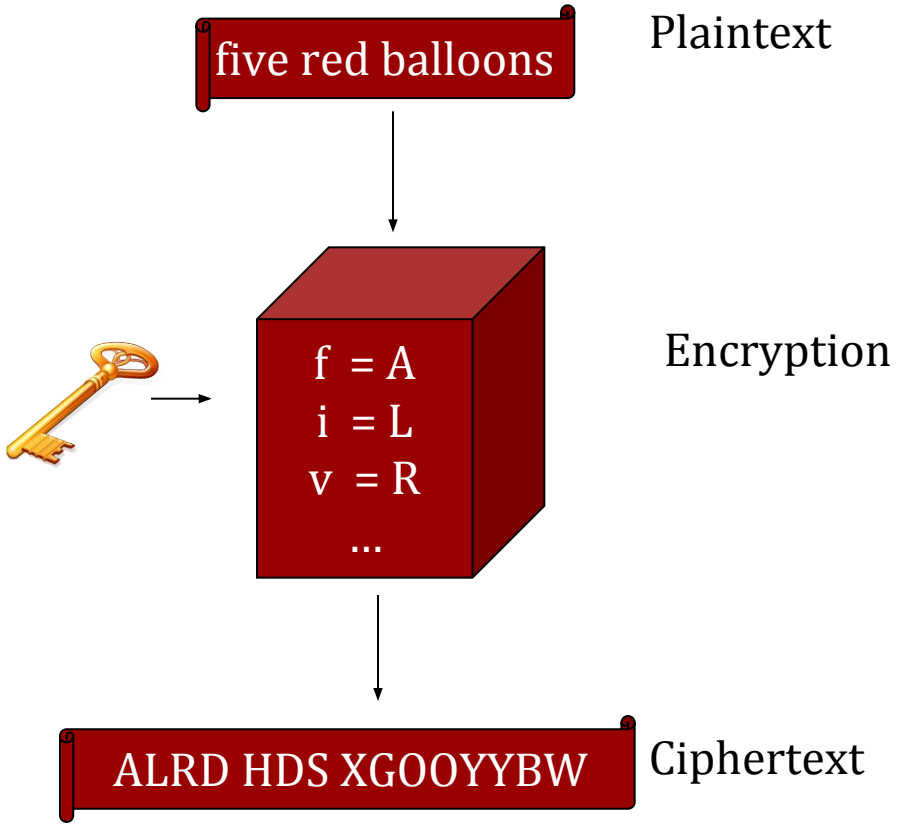
- Rather than having a fixed shift, change every plaintext letter to an arbitrary ciphertext letter

Plaintext	Ciphertext
a	G
b	X
c	N
d	S
e	D
…	…
z	Q

The Substitution Cipher cont.

Key =

a	G	n	B
b	X	o	Y
c	N	p	Z
d	S	q	P
e	D	r	H
f	A	s	W
g	F	t	I
h	V	u	J
i	L	v	R
j	M	w	U
k	C	x	K
l	O	y	T
m	E	z	Q



The Substitution Cipher cont.

- To decrypt we just look up the ciphertext letter in the table and then write down the matching plaintext letter

- How many keys do we have now?
  - A key is just a permutation of the letters of the alphabet
  - There are 26! permutations
    - 403291461126605635584000000

Breaking the Substitution Cipher

- If we could do 1 000 000 calculations per second, it would take over 12.7×10^{12} years to try all possible keys
  - The universe is about 13.7×10^{9} years old

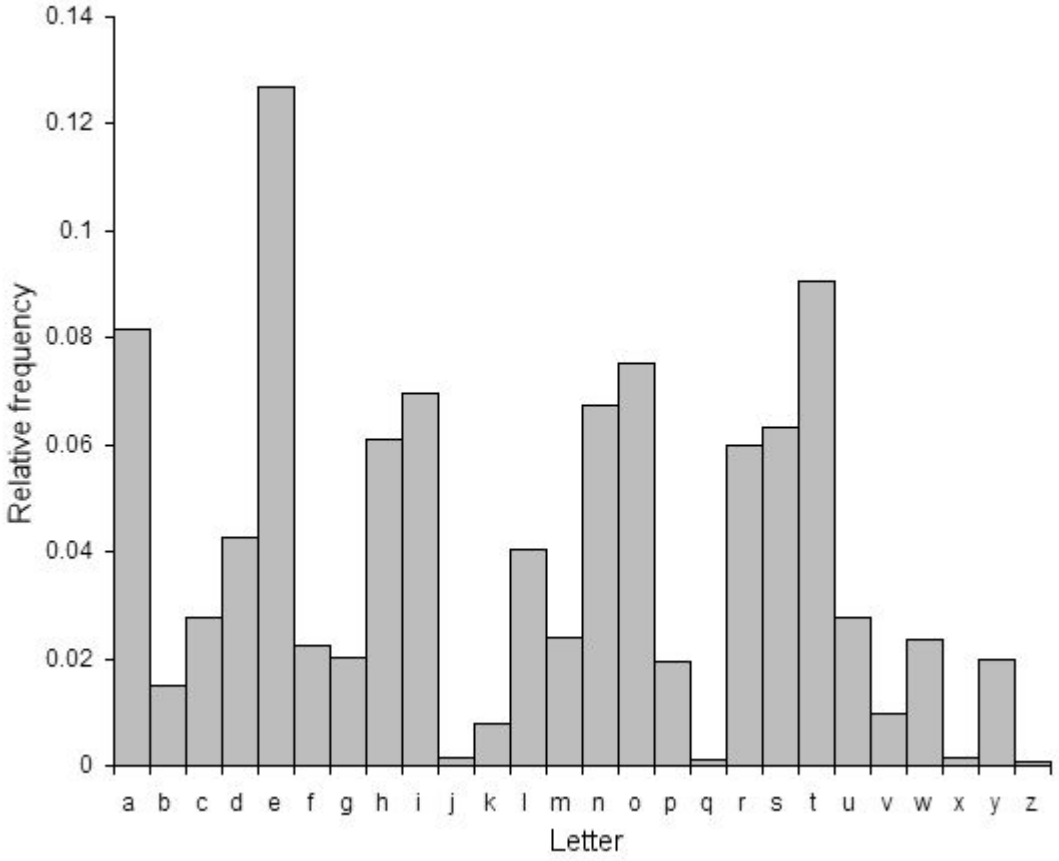
- Trying all possible keys is probably not going to work

Frequency Analysis

- In English (or any language) certain letters are used more often than others
- If we look at a ciphertext, certain ciphertext letters are going to appear more often than others
- It would be a good guess that the letters that occur most often in the ciphertext are actually the most common English letters

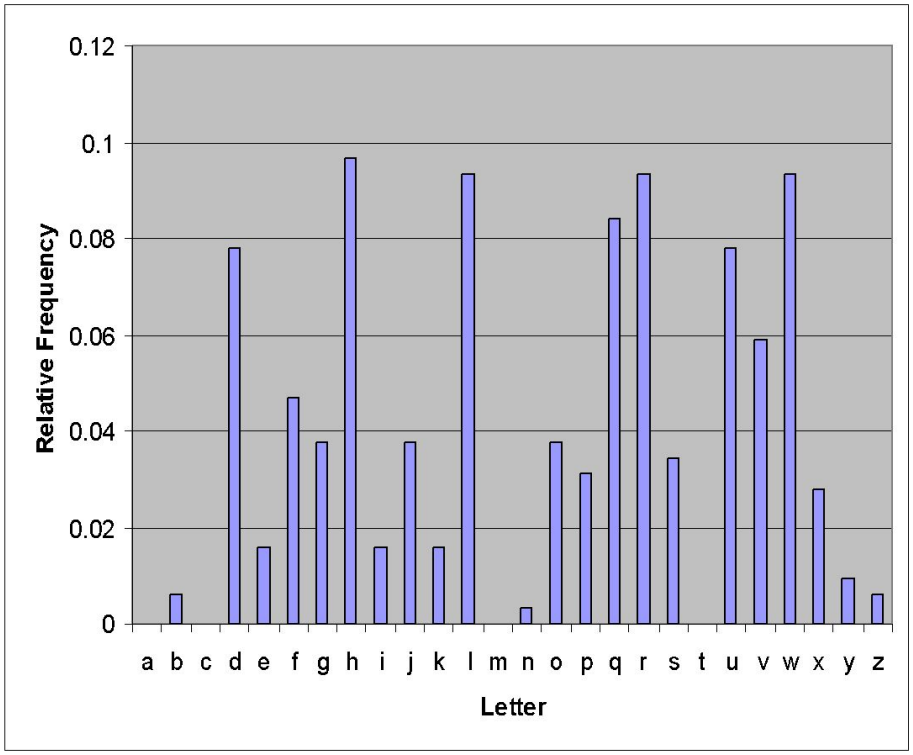
Letter Frequency

- This is the letter frequency for English
- The most common letter is ‘e’ by a large margin, followed by ‘t’, ‘a’, and ‘o’
- ‘J’, ‘q’, ‘x’, and ‘z’ hardly occur at all

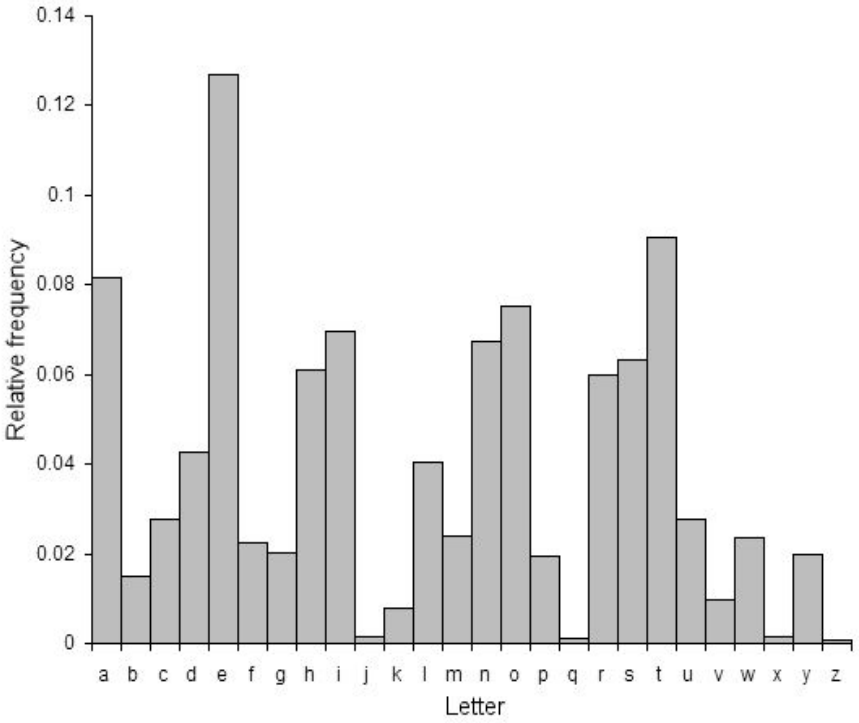


Frequency Analysis in Practice

- Suppose this is our ciphertext
  - dq lqwurgxfwlrq wr frpsxwlqj surylglqj d eurdg vxuyhb ri wkh glvflsolqh dqg dq lqwurgxfwlrq wr surjudpplqj. vxuyhb wrslfv zloo eh fkrvhq iurp: ruljlqv ri frpsxwhuv, gdwd uhsuhvhqwdwlrq dqg vwrudjh, errohdq dojheud, gljlwdo orjlf jdwhv, frpsxwhu dufklwhfwxuh, dvvhpeohuv dqg frpslohuv, rshudwlqj vbvwhpv, qhwzrunv dqg wkh lqwhuqhw, wkhrulhv ri frpsxwdwlrq, dqg duwlilfldo lqwhooljhqfh.



Ciphertext distribution



English distribution

In our ciphertext we have one letter that occurs more often than any other (h), and 6 that occur a good deal more than any others (d, l, q, r, u, and w)

There is a good chance that h corresponds to e, and d, l, q, r, u, and w correspond to the 6 next most common English letters

Caesar Cipher: c = m + 3

A B C D E F G H I J

K L M N O P Q R

S T U V W X Y Z



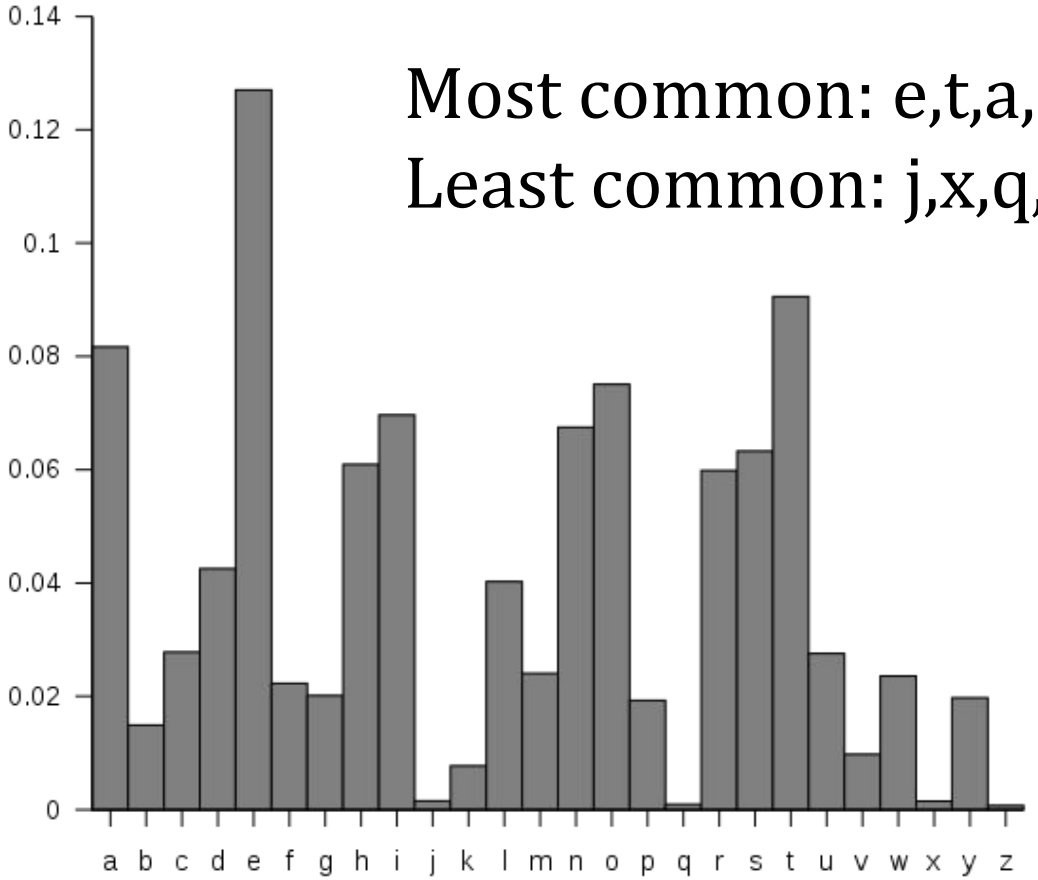
Julius Caesar
100 BC- 44 BC

How would you *attack* messages encrypted with a substitution cipher?

Attacking Substitution Ciphers

Trick 1:
Word Frequency

Trick 2:
Letter Frequency

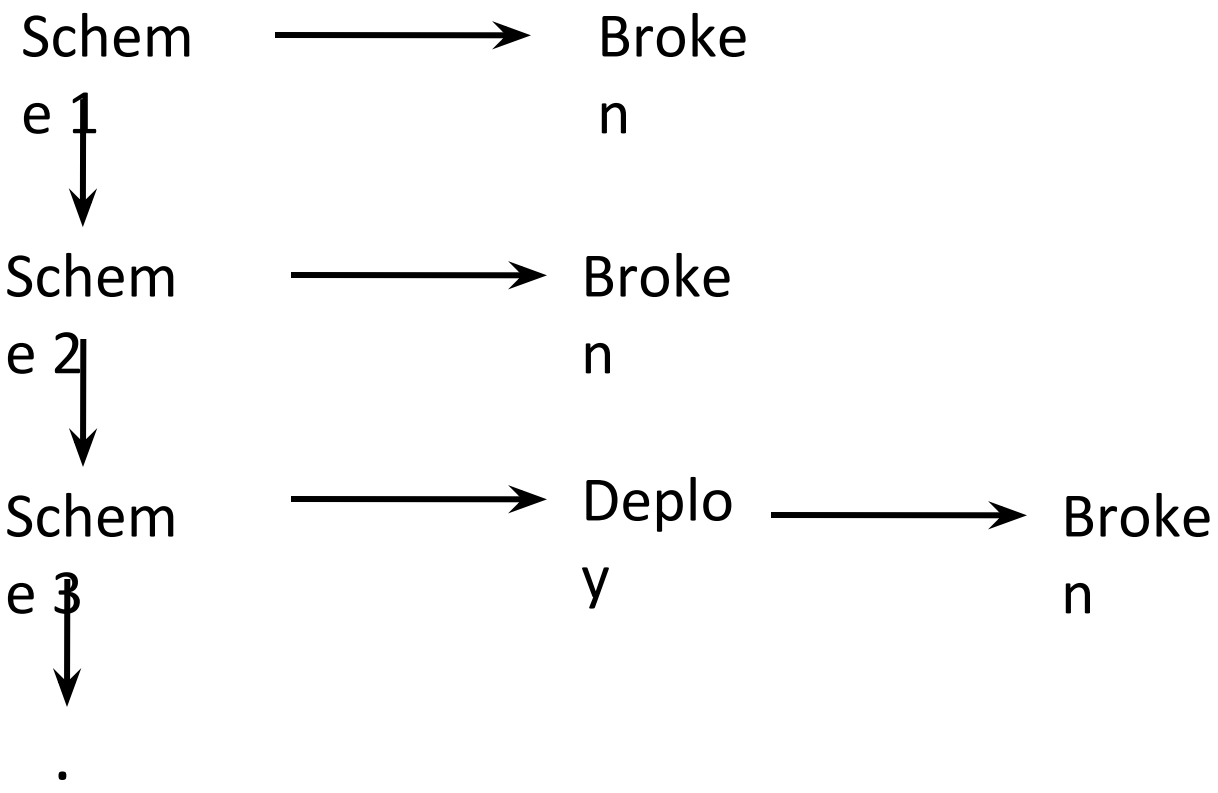


Most common: e,t,a,o,i,n
Least common: j,x,q,z

image source: wikipedia

Jvl mlwclk yr jvl owmwez twp yusl w zyduo pjdcluj mqil zydkplmr. Hdj jvlz tykilc vwkc jy mlwku jvl wkj yr vwsiquo, tvqsv vlmflc mlwc jvlg jy oklwjulpp. Zyd vwnl jvl fyjlujqwm jy cy jvl pwgl. Zydk plsklj fwpptykc qp: JYWPJ

Classical Approach: Iterated Design



No way to say anything is secure
(and you may not know when broken)

Frequency Analysis cont.

- If we replace ‘e’ with ‘h’ and the 6 next most common letters with their matches, the ciphertext becomes
  - an intro???tion to ?o?p?tin? pro?i?in? a ?roa? ??r?e? o? t?e ?i??ip?ine an? an intro???tion to pro?ra??in?. ??r?e? topi?? ?i?? ?e ??o?en ?ro?: ori?in? o? ?o?p?ter?, ?ata repre?entation an? ?tora?e, ?oo?ean a??e?ra, ?i?ita? ?o?i? ?ate?, ?o?p?ter ar??ite?t?re, a??e???er? an? ?o?pi?er?, operatin? ???te??, net?or?? an? t?e internet, t?eorie? o? ?o?p?tation, an? arti?i?ia? inte??i?en?e.

List of classical ciphers

1. Substitution Ciphers

Ciphers that replace each symbol with another symbol.

Monoalphabetic Substitution

- Caesar Cipher
- Shift Cipher
- Affine Cipher
- Atbash Cipher
- Keyword Cipher
- Pigpen Cipher
- Bacon's Cipher
- Monoalphabetic Substitution (General)

Polyalphabetic Substitution

- Vigenère Cipher
- Beaufort Cipher
- Variant Beaufort Cipher

2. Transposition Ciphers

Ciphers that rearrange the order of characters without altering them.

- Rail Fence Cipher
- Columnar Transposition
- Route Cipher
- Scytale Cipher (Ancient Greek)
- Double Transposition Cipher
- Myszkowski Transposition
- Permutation Cipher (General)

3. Stream-Based Classical Systems

Early forms of stream or pseudo-stream encryption:

- One-Time Pad (classical form)
- Vernam Cipher

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4. Mechanical / Electro-mechanical Ciphers

Pre-modern but before modern cryptography:

- Enigma Machine
- Hebern Rotor Machine
- Hagelin Machines (M-209, CX-52)
- SIGABA
- Lorenz SZ40/42 (Tunny)

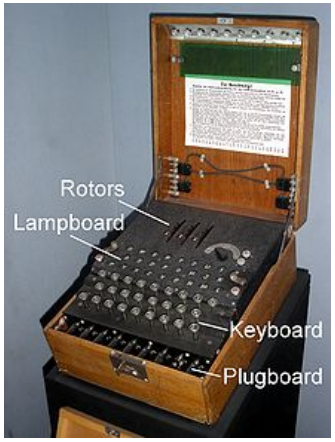
Machine Ciphers

Enigma Machine?

- An Enigma machine is an electro-mechanical rotor cipher machines used for the encryption and decryption of secret massages.

- Rotors (moving substitution wheels)
- Reflector (ensures symmetric encryption)
- Plugboard (letter pair swapping)
- Lampboard (ciphertext output)

- Keyboard input encrypted through rotors
- Rotors perform shifting substitution
- Reflector returns signal back
- Plugboard adds extra permutations



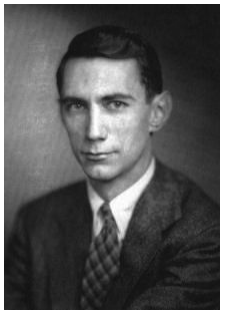




Modern Cryptography

- Frequency analysis was the best cryptanalysis until the invention of computers.
- In World War II, the Nazi military employed an encryption scheme that addressed the weakness of substitution ciphers.
- The scheme, implemented by typewriter sized devices known as Enigma machines, gave the Nazis a tactical advantage that greatly contributed to their early success in the war.

Claude Shannon



- Formally define:
  - *security goals*
  - *adversarial models*
  - *security of system wrt goals*

- Beyond iterated design: Proof!

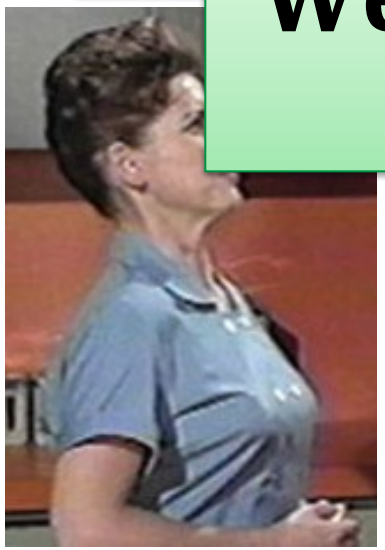
Motivation

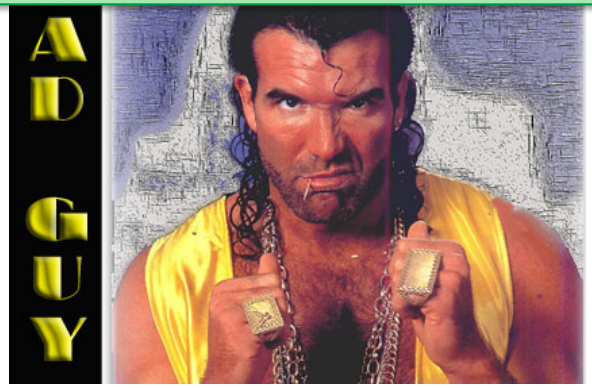
- Alice wants to Communicate with Bob
- Alice and Bob are unable to communicate on the untrusted network

Ali

Bob

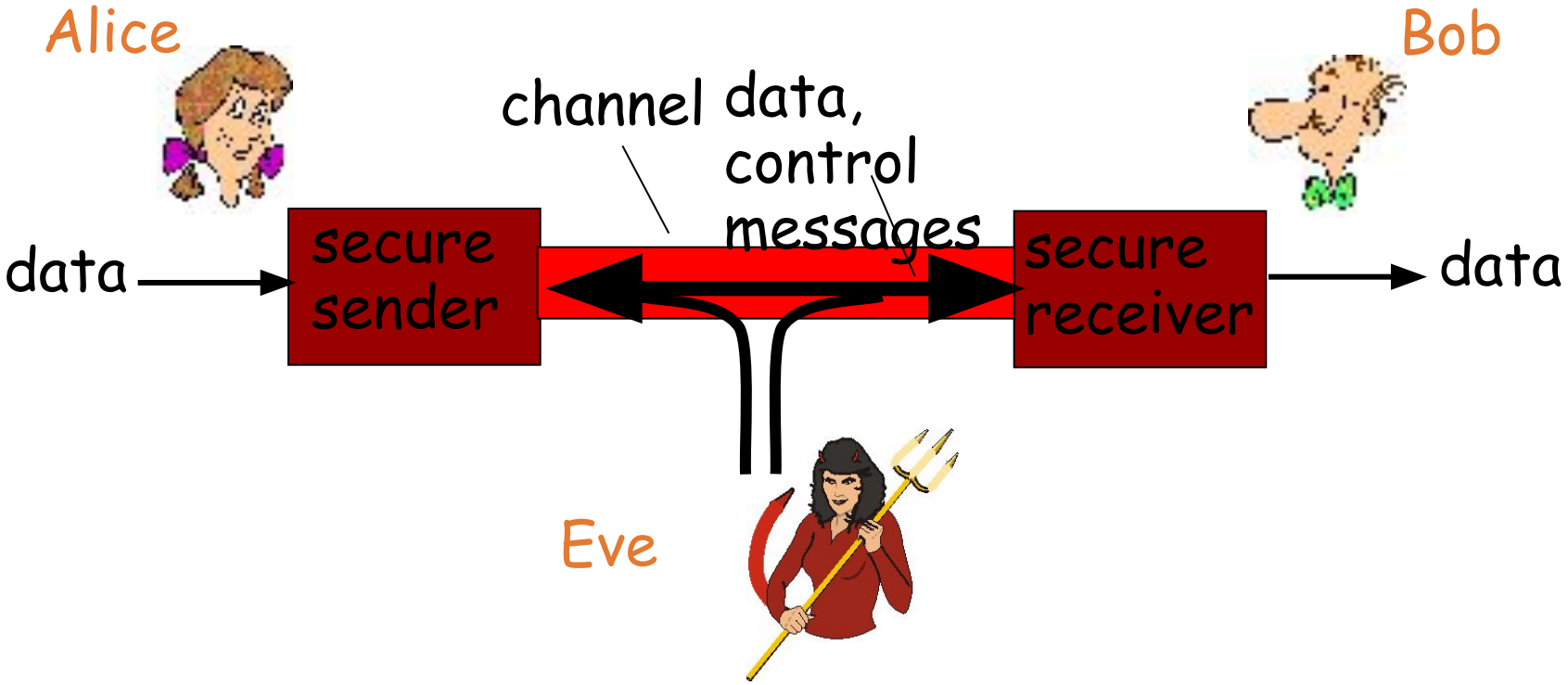
We need Cryptography







Motivation


Alice
Bob
channel
data,
control
messages
data
secure
sender
secure
receiver
data
Eve

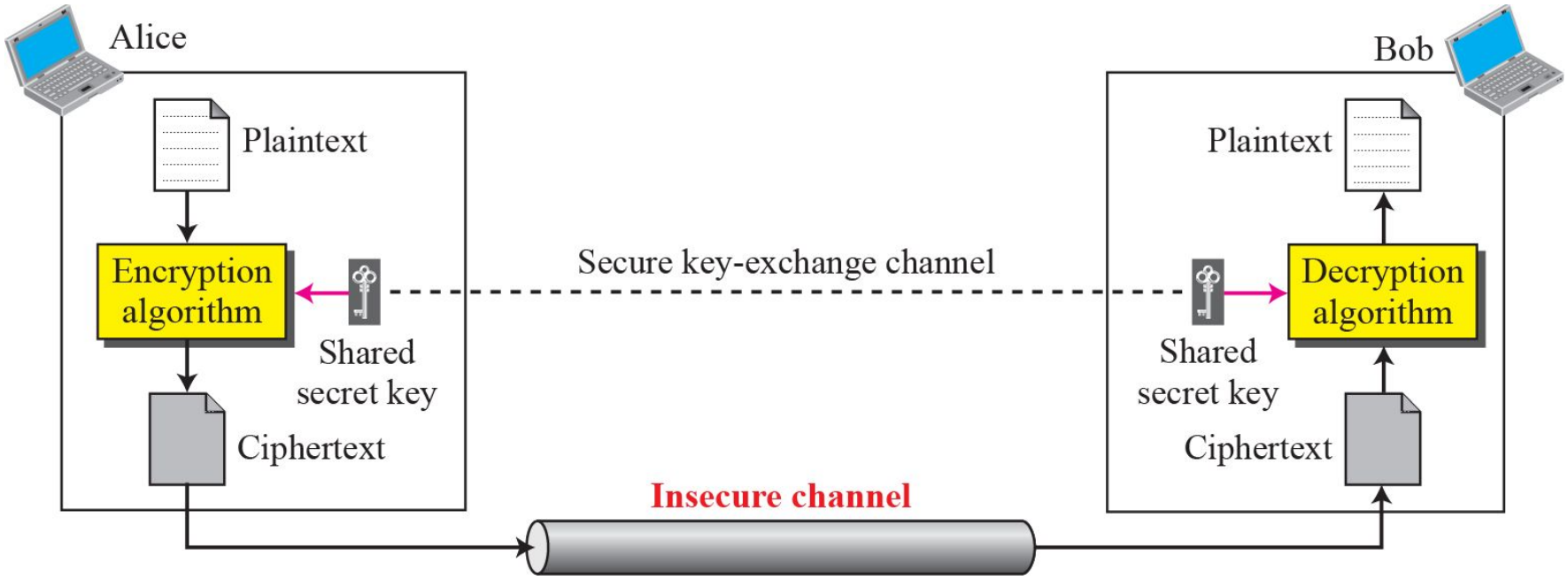
Outline

- Introduction
- Security and Privacy
- Cryptography
  - Secret Key Crypto (Symmetric)
  - Public Key Crypto (Asymmetric)
- Key Management
- Cryptanalysis

Basic Terminology

- Plain-text - the original message
- Cipher-text - the coded message
- Key - info used in cipher known only to sender/receiver
- Encipher (encrypt) - converting plain-text to cipher-text
- Decipher (decrypt) – converting cipher-text to plain-text
- Cryptography- study of encryption/decryption principles /methods
- Cryptanalysis (code breaking) - the study of principles/ methods of deciphering cipher-text without knowing key
- Cryptology - the field of both cryptography and cryptanalysis

General idea of traditional cipher



Goals of Crypto

- Confidentiality is the concealment of information or resources.
  - E.g., only sender, intended receiver should "understand" message contents

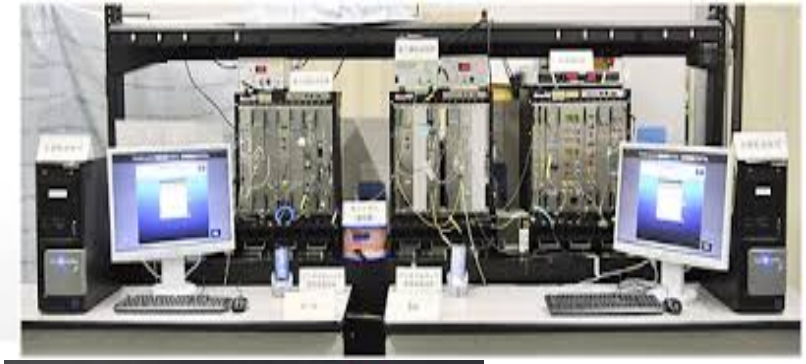
- Authenticity is the identification and assurance of the origin of information.

- Integrity refers to the trustworthiness of data or resources in terms of preventing improper and unauthorized changes.

- Availability refers to the ability to use the information or resource desired.

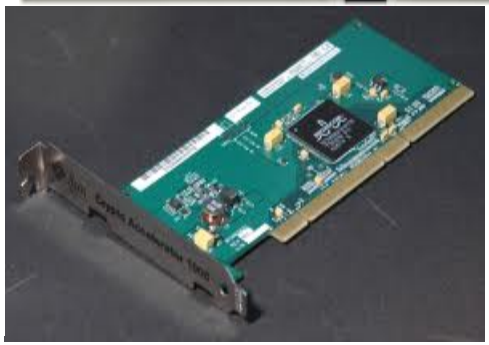
Crypto in Real World







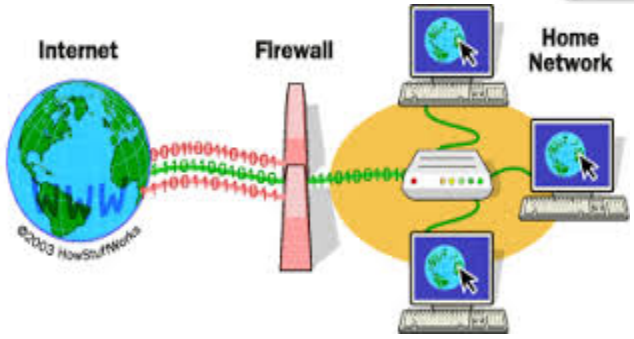














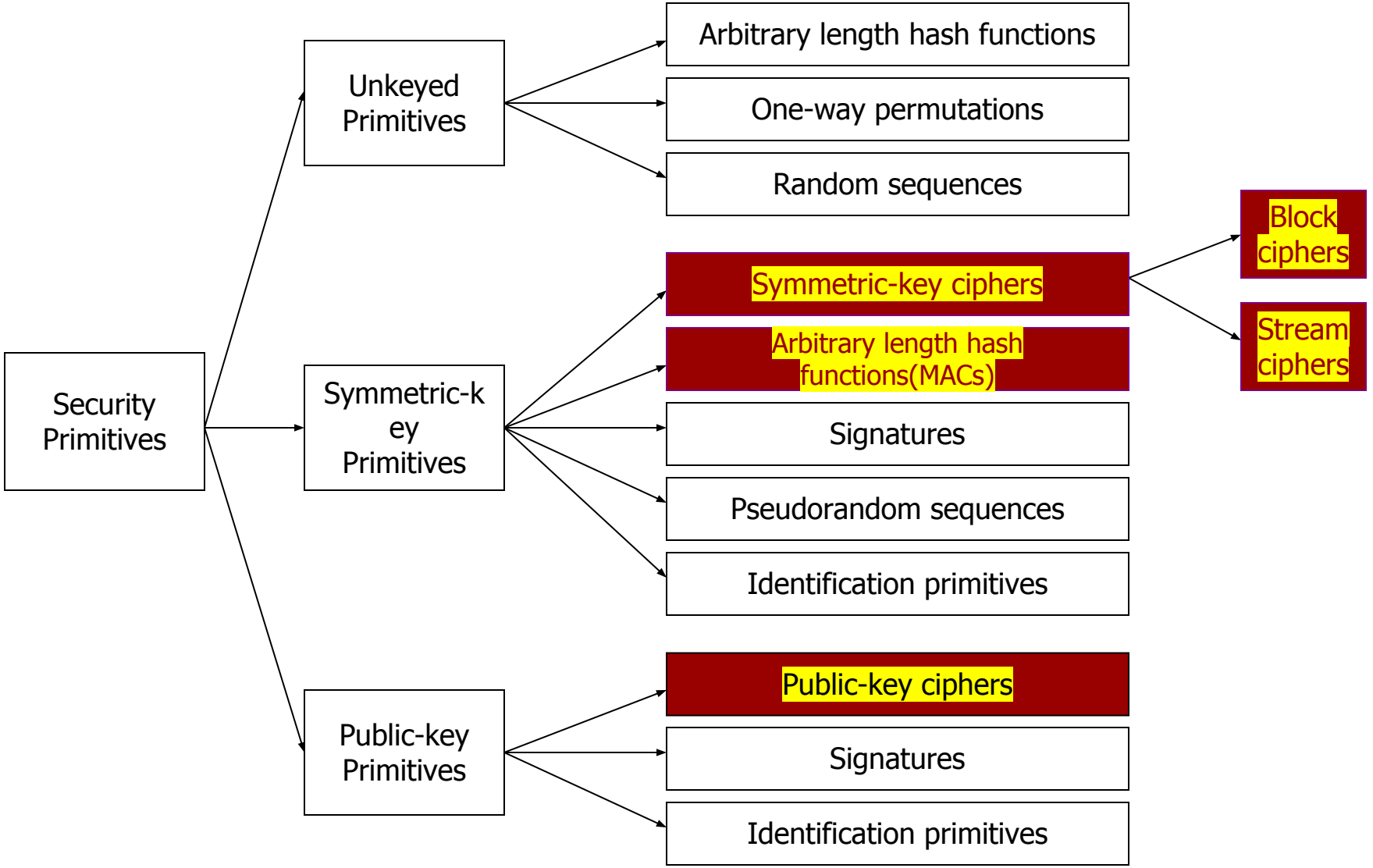
Classify Security Attacks as

- **Passive attacks** - eavesdropping on, or monitoring of, transmissions to:
  - obtain message contents, or
  - monitor traffic flows
- **Active attacks** – modification of data stream to:
  - masquerade of one entity as some other
  - replay previous messages
  - modify messages in transit
  - denial of service

Security Mechanisms (X.800)

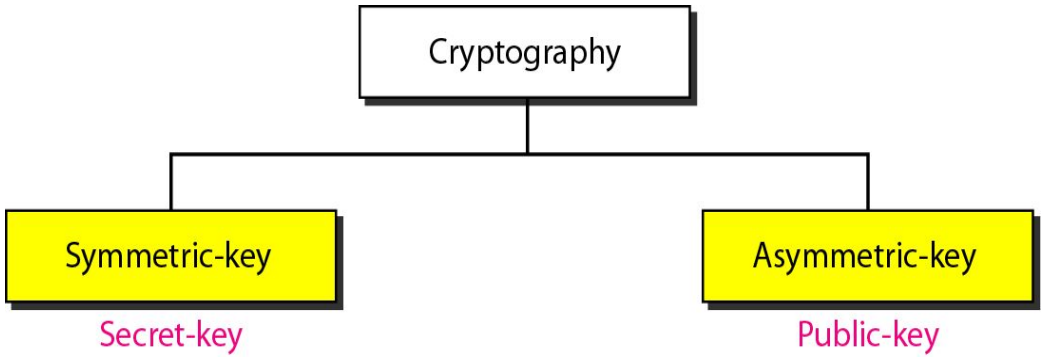
- Encipherment
- Digital signatures
- Access controls
- Authentication exchange
- Traffic padding

Cryptographic Primitives

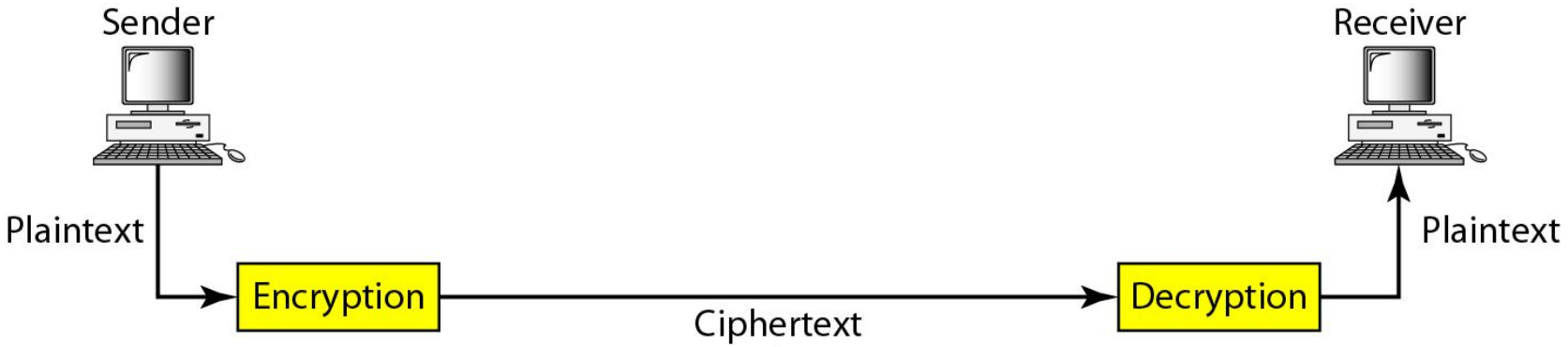


Cryptography components

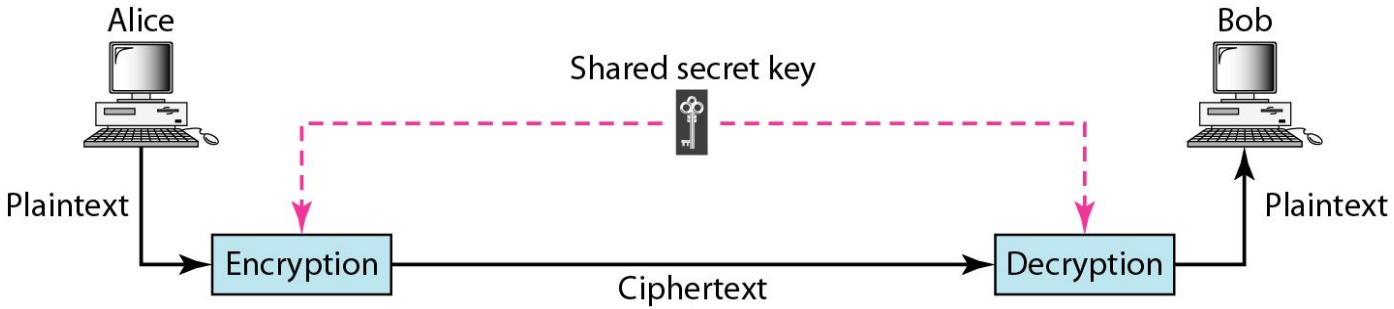
Types



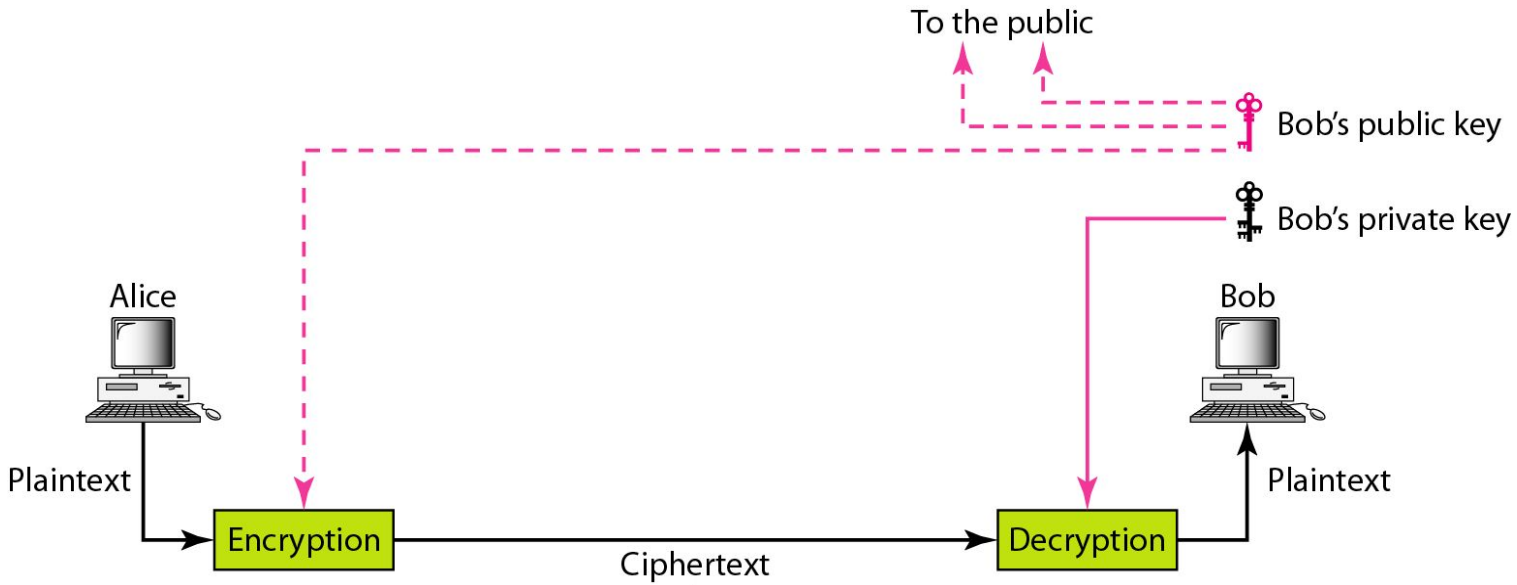
Cipher



Symmetric-key cryptography



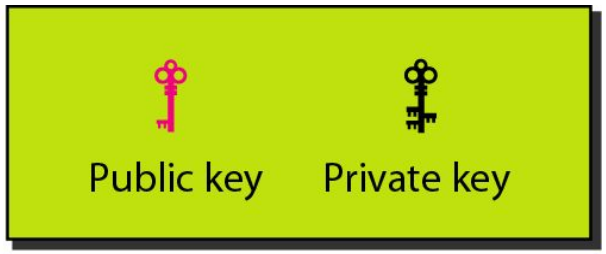
Asymmetric-key cryptography



Keys used in cryptography

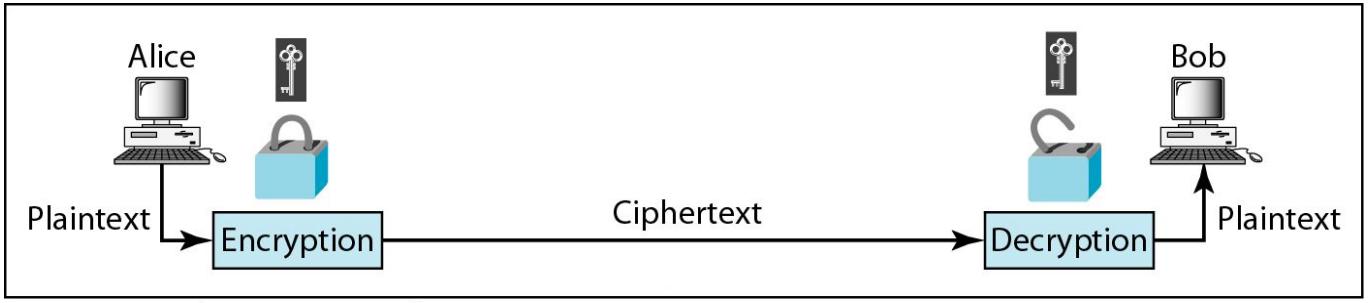


Symmetric-key cryptography

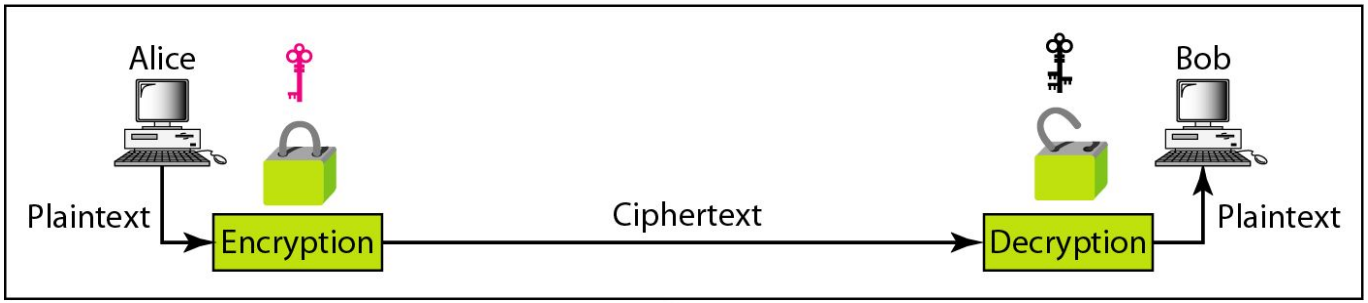


Asymmetric-key cryptography

Cryptosystems



a. Symmetric-key cryptography



b. Asymmetric-key cryptography

Cryptosystems

- Private Key Crypto (Symmetric key)
  - Ex: Stream ciphers (A5/1, A5/3, RC4 etc.)
  - Block ciphers (DES, AES, KASUMI etc)

- Public key Crypto (Asymmetric key)
  - Ex: RSA, Diffie-Hellman, Elgamal, Rabin etc..

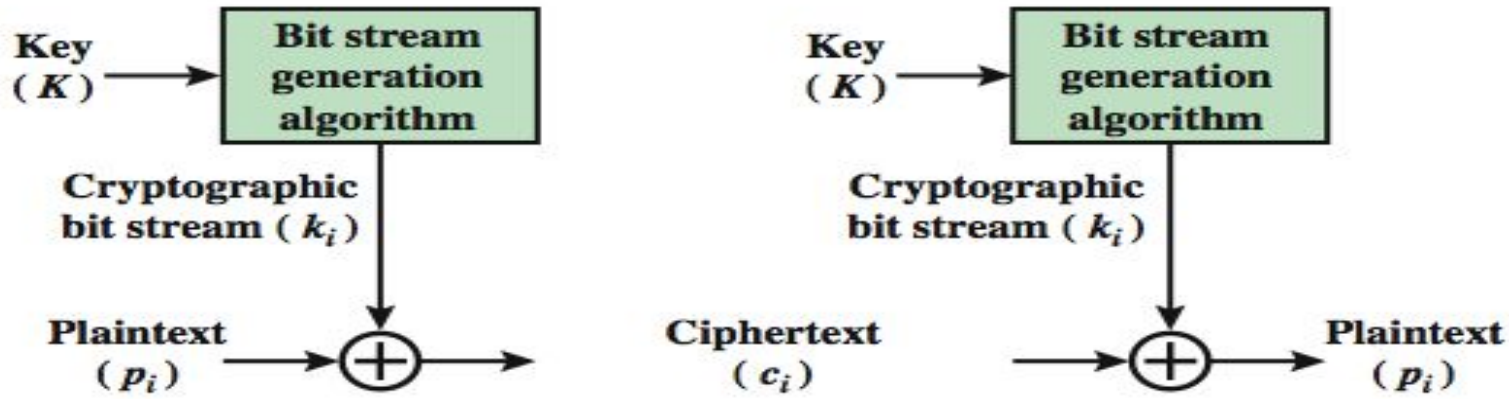
Stream Ciphers

(Symmetric key Cryptography)

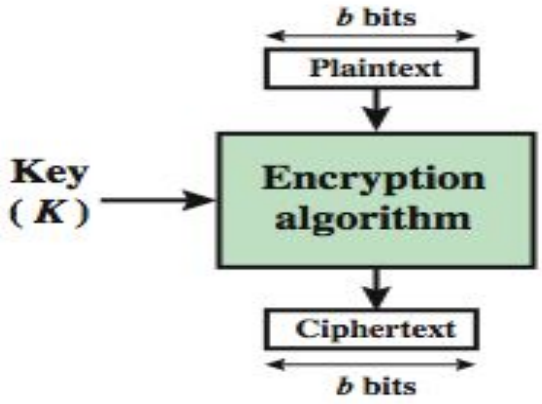
Basics

- Notation
  - Secret key K
  - Encryption function $E_K(P)$
  - Decryption function $D_K(C)$
  - Plaintext length typically the same as ciphertext length
  - Encryption and decryption are one-one mapping functions on the set of all n-bit arrays
- Efficiency
  - functions E_K and D_K should have efficient algorithms
- Consistency
  - Decrypting the ciphertext yields the plaintext
  - $D_K(E_K(P)) = P$

Process of Symmetric key ciphers



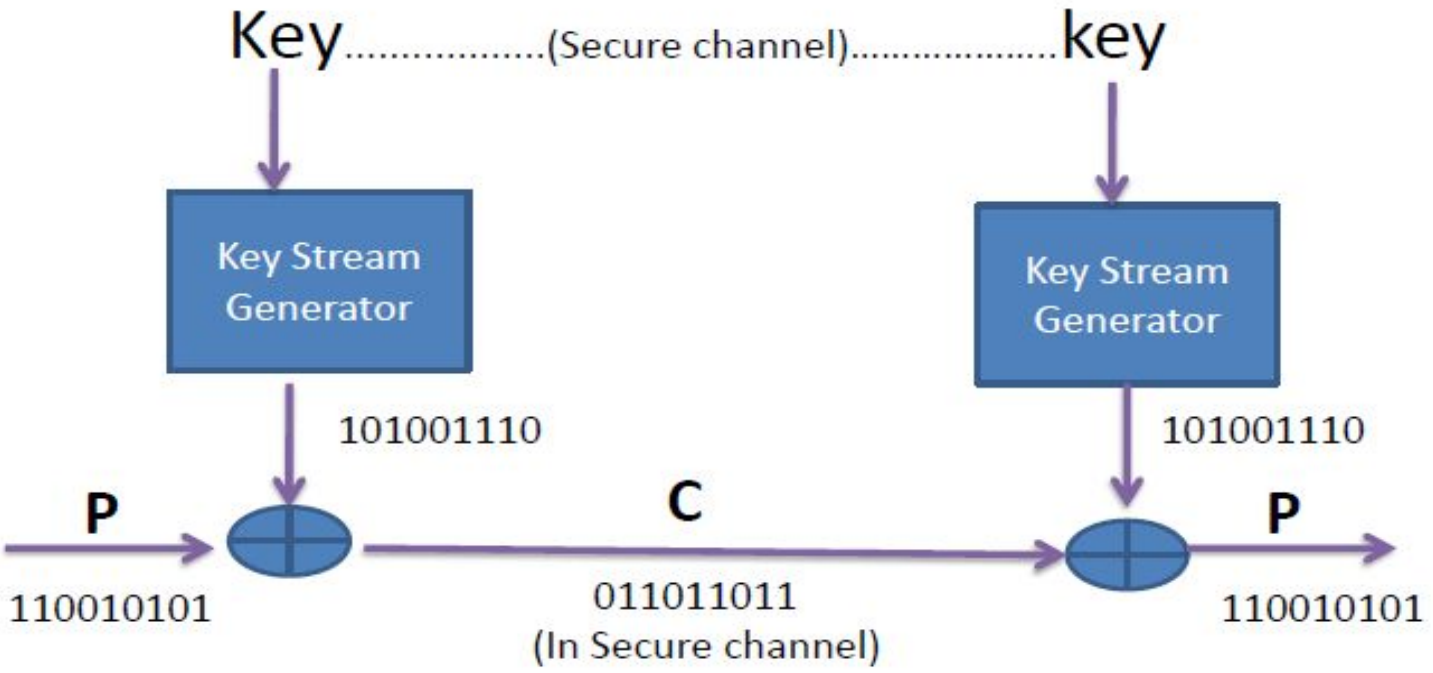
(a) Stream Cipher Using Algorithmic Bit Stream Generator



(b) Block Cipher

Stream Cipher

- Simple Stream Cipher:



Stream Cipher

- Stream ciphers that encrypt an arbitrary stream of data.
  - ❑ Depending on the cipher, the data may consist of a stream of bits or a stream of bytes.
  - ❑ Each plaintext bit (or byte) encrypts to a different cipher text bit (or byte) depending on what data occurred earlier in the stream.
  - ❑ It is possible to use a block cipher to implement a stream cipher.

- Stream Ciphers

  Basic idea: replace the random key in the one-time pad by a pseudo-random sequence, generated by a cryptographic pseudo-random generator (PRG) that is 'seeded' with the key.

PRG / KSG

- **A pseudo random generator is a polynomial time computable function G : $\{0,1\}^n$ → $\{0,1\}^m$(on input of length n it produces an output of length m), where $m = n^c$; $c > 1$, which satisfies:**
  - **The output of G is polynomial time indistinguishable from truly random strings of length m.**

- **Notice that the output of such G cannot be truly random!**

PRG and SEED

- PRG requirements:
- Randomness

  – Uniformity, scalability, consistency
- Unpredictability

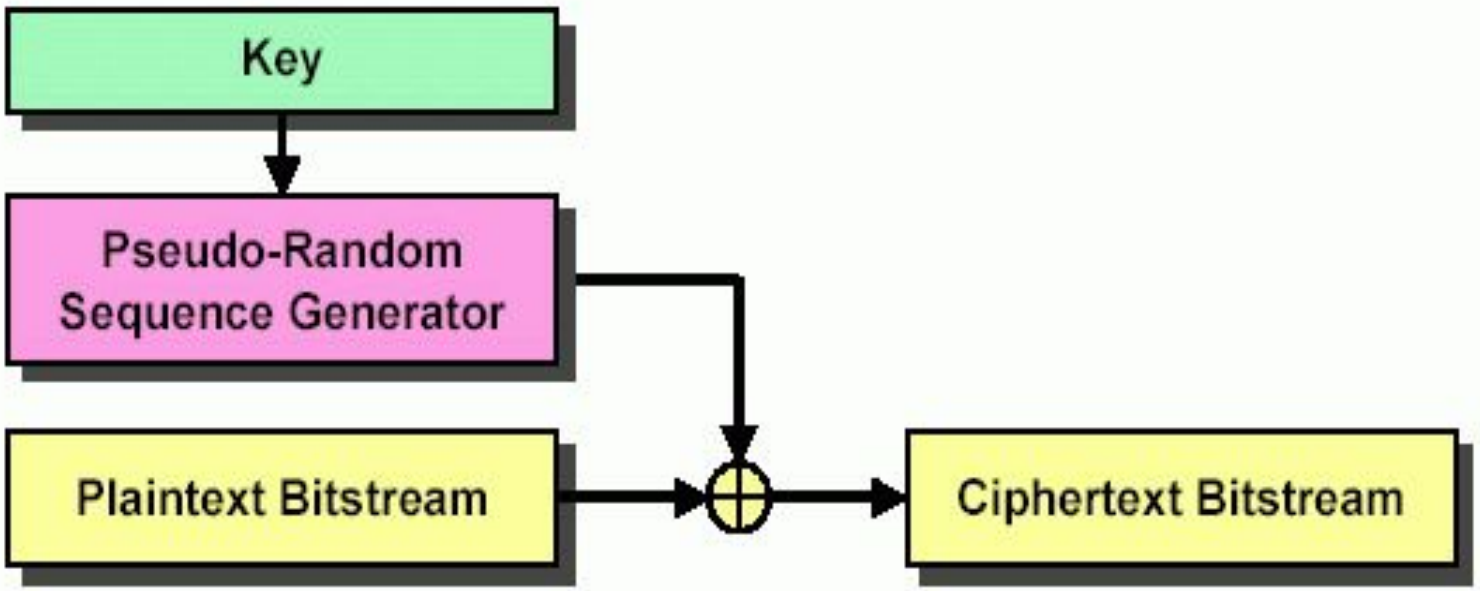
  – Cannot determine what next bit will be despite knowledge of the algorithm and all previous bits.
- Unreproducable

  – Cannot be reliably reproduced.

Characteristics of the seed:

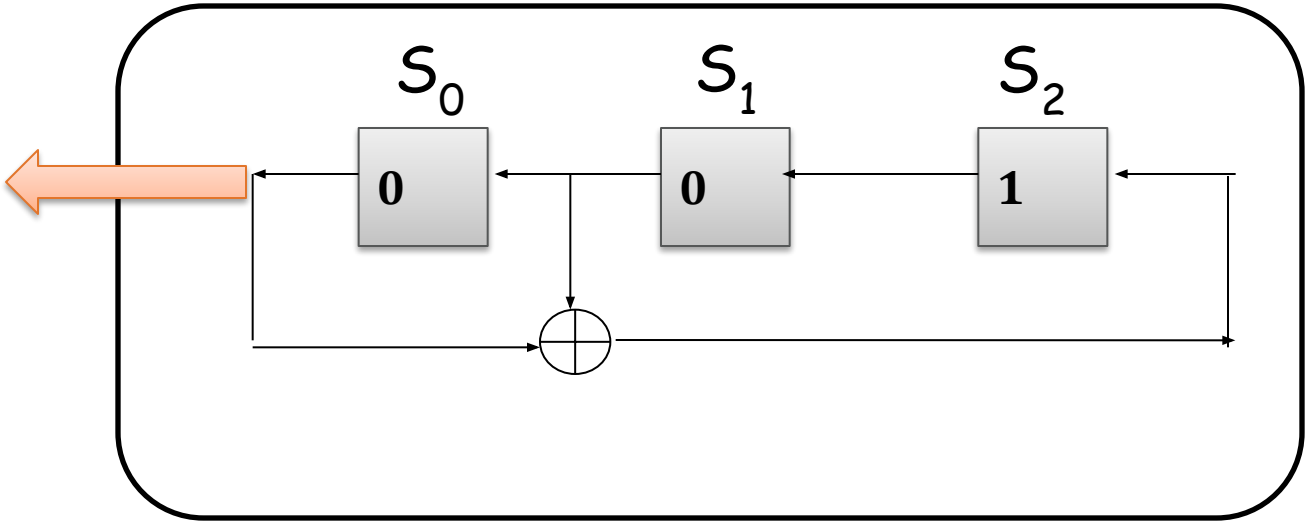
- ❑ Must be kept secret
- ❑ If known, adversary can determine output
- ❑ Must be random or pseudorandom number

Stream cipher: Process



Plaintext Stream	1	1	1	1	1	1	1	1	0	0	0	0	0	0 ...
Pseudo-Random Stream	1	0	0	1	1	0	1	0	1	1	0	1	0	0 ...
Ciphertext Stream	0	1	1	0	0	1	0	1	1	1	0	1	0	0 ...

Linear Feedback Shift Register



$s_{t+2} = s_{t+1} + s_t$

S_0	S_1	S_2
0	0	1
0	1	0
1	0	1
0	1	1
1	1	1
1	1	0
1	0	0
---	---	---
0	0	1
0	1	0

Initial fill determines the sequence of states

Generates a periodic sequence

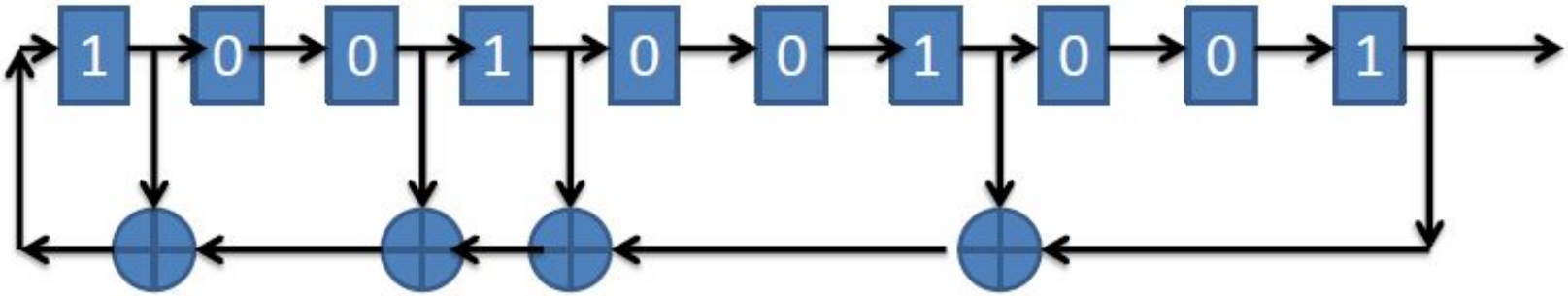
...0010111...

Maximal period $2^3-1=7$

Irreducibility

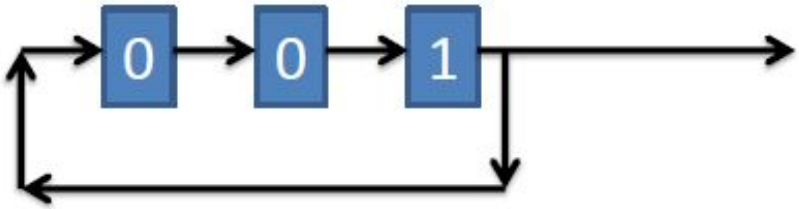
Two LFSRs Produce the same sequence.

$P(X) = 1+X+X^3+X^4+X^7+X^{10}$



$P(X) = 1+X+X^3+X^4+X^7+X^{10} = (1+X+X^7)(1+X^3)$

$P(X) = 1+X^3$



Periodicity

- For an initial state of all zeros the o/p sequence will be the zero sequence.
- For a non zero initial state the o/p sequence must be eventually periodic.
- In fact there are 2^L -1 non zero states and so an LFSR, for all non zero initial states, produces an output stream whose period is of length exactly 2^L -1. It is possible when the feedback polynomial is Primitive.

Perfectness

- (**Shannon secrecy). (M, K, Gen, Enc, Dec)** is said to be a private-key encryption scheme that is Shannon-secret with respect to the distribution D over the message space M

  if for all m0 $\in$ M and for all c,

  Pr [k $\leftarrow$ Gen; m $\leftarrow$ D : m = m0 | Enc**k**(m) = c] = Pr [m $\leftarrow$ D : m = m0]

- **Definition 11.2 (Perfect Secrecy).** A tuple (M, K, Gen, Enc, Dec) is said to be a private-key encryption scheme that is perfectly secret if for all m1 and m2 in M, and for all c,

  Pr[k $\leftarrow$ Gen : Enc**k**(m1) = c] = Pr[k $\leftarrow$ Gen : Enc**k**(m2) = c].

- **Unconditional Security:** A cryptosystem is unconditionally or information theoretically secure if it cannot be broken even with infinite computational resources

ENCRYPTION



Message 1

Central to the growth of e-commerce and e-governance is the issue of trust in electronic environment.

Encrypted Message 1

9a468[illegible]4335be49f0b9cab28d755aaa9cd985
71b275bb[illegible]405e6931e856ca3e5e569e
dd135285482

Same Key
SYMMETRIC

Message 2

The Internet knows no g[illegible]
It has redefined time and space. [illegible] in computer and telecommunication technologies have led to the explosive growth of the Internet. This in turn is affecting the methods of communication, work, study, education, interaction, leisure, health, governance, trade and commerce.

Encrypted Message 2

a[illegible]cb61a770f947ca856cd675463f1c95a
9a2[illegible]71f80830c87f5715f5f5933497[illegible]
dd7e[illegible]
7c398[illegible]

Different Keys
[Keys of a pair – Public and Private]
ASYMMETRIC
[PKI]

DECRYPTION



Encrypted Message 1

9a46894335be49f0b9cab28d755aaa9cd98571b
275bbb0adb405e6931e856ca3e5e569edd13528
5482

Message 1

Central to the growth of e-commerce and e-governance is the issue of trust in electronic environment.

Encrypted Message 2

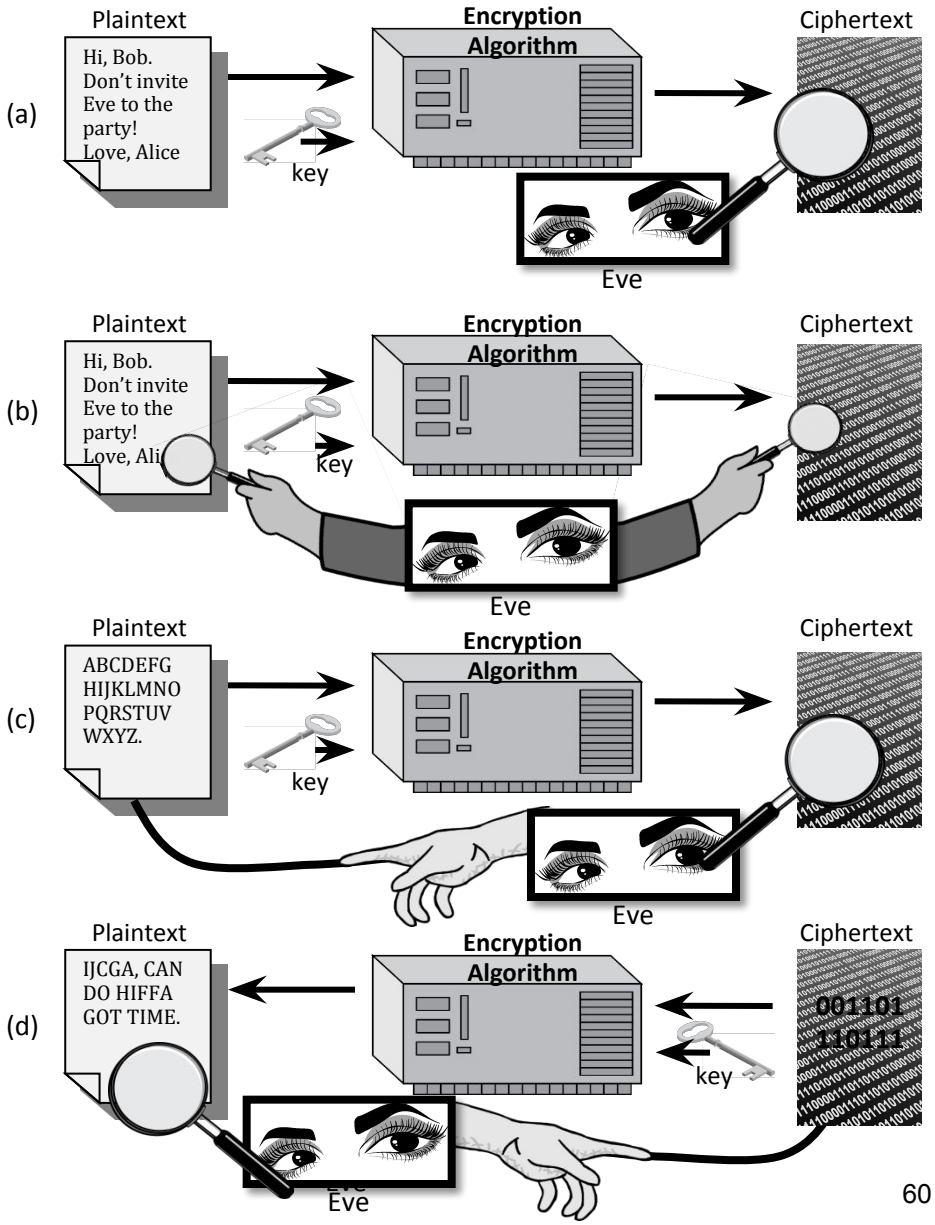
a520eecb61a770f947ca856cd675463f1c95a9a2b
8d4e6a71f80830c87f5715f5f59334978dd7e97da
0707b48a1138d77ced56feba2b467c398683c7db
eb86b854f120606a7ae1ed934f5703672adab0d7
be66dccde1a763c736cb9001d0731d541106f50b
b7e54240c40ba780b7a553bea570b99c9ab3df13
d75f8ccfdddeaaf3a749fd1411

Message 2

The Internet knows no geographical boundaries. It has redefined time and space. Advances in computer and telecommunication technologies have led to the explosive growth of the Internet. This in turn is affecting the methods of communication, work, study, education, interaction, leisure, health, governance, trade and commerce.

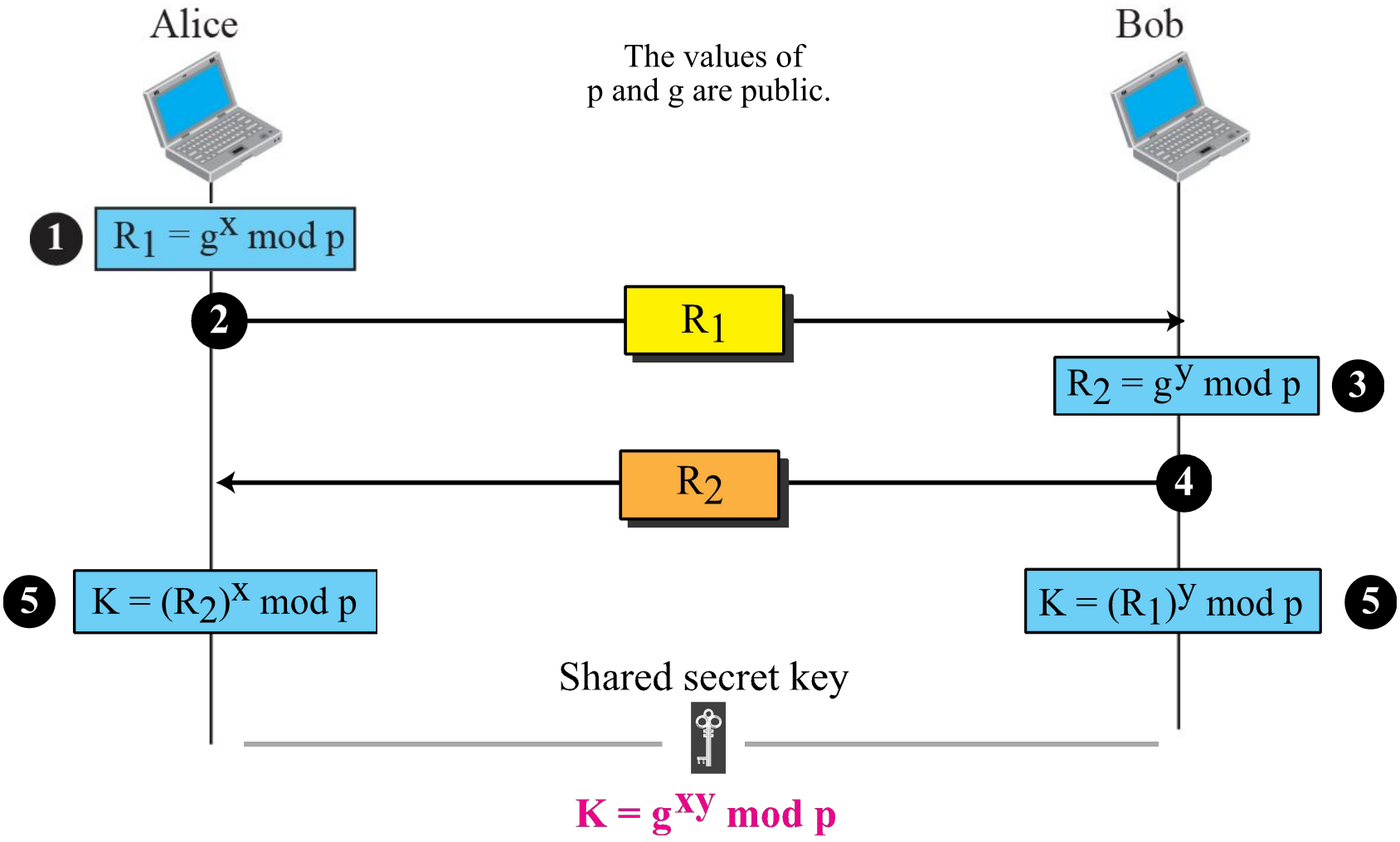
Attacks

- **Attacker may have**
  - a) **collection of ciphertexts (ciphertext only attack)**
  - b) **collection of plaintext/ciphertext pairs (known plaintext attack)**
  - c) **collection of plaintext/ciphertext pairs for plaintexts selected by the attacker (chosen plaintext attack)**
  - d) **collection of plaintext/ciphertext pairs for ciphertexts selected by the attacker (chosen ciphertext attack)**



Key Exchange Protocol

Diffie-Hellman Key Exchange



Example

- Alice and Bob get public numbers
  - P = 23, G = 9
- Alice and Bob compute public values
  - $X = 9^4 \bmod 23 = 6561 \bmod 23 = 6$
  - $Y = 9^3 \bmod 23 = 729 \bmod 23 = 16$
- Alice and Bob exchange public numbers

Example

- Alice and Bob compute symmetric keys
  - $k_a = y^a \bmod p = 16^4 \bmod 23 = 9$
  - $k_b = x^b \bmod p = 6^3 \bmod 23 = 9$
- Alice and Bob now can talk securely!



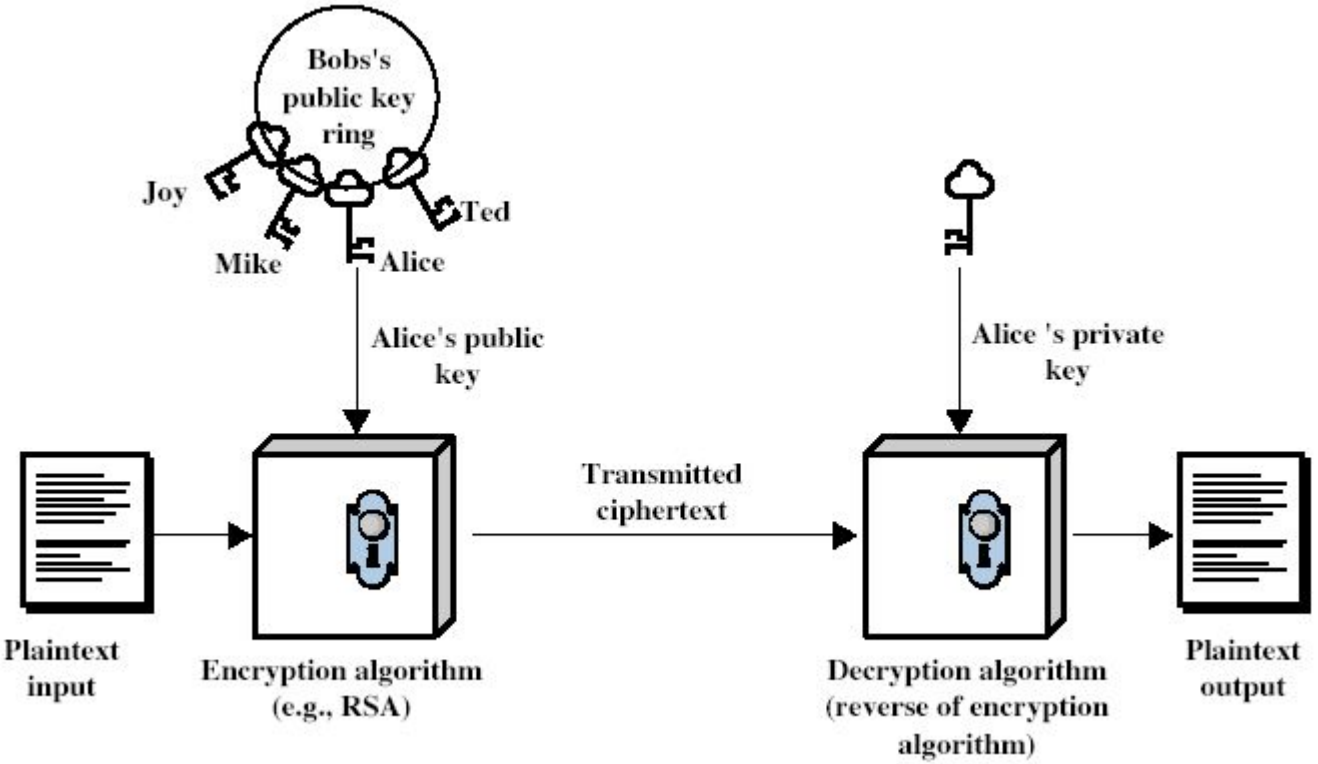


Public Key Cryptography

Asymmetric-Key Cryptography

- **public-key/two-key/asymmetric** cryptography involves the use of **two** keys:
  - a **public-key**, which may be known by anybody, and can be used to **encrypt messages**, and **verify signatures**
  - a **private-key**, known only to the recipient, used to **decrypt messages**, and **sign** (create) **signatures**
- is **asymmetric** because
  - those who encrypt messages or verify signatures **cannot** decrypt messages or create signatures

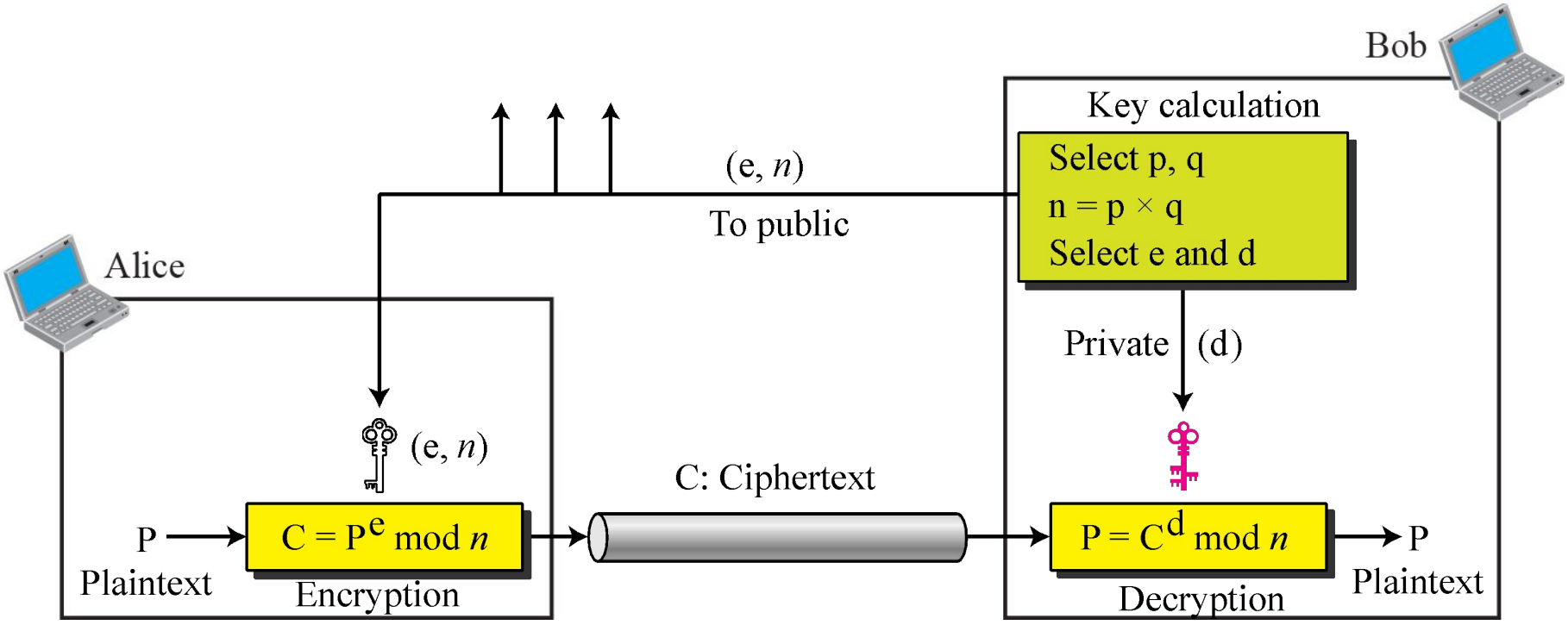
Public-Key Cryptography



RSA algorithm

- by Rivest, Shamir & Adleman of MIT in 1977
- best known & widely used public-key scheme
- based on exponentiation in a finite (Galois) field over integers modulo a prime
  - nb. exponentiation takes $O((\log n)^3)$ operations (easy)
- uses large integers (eg. 1024 bits)
- security due to cost of factoring large numbers
  - nb. factorization takes $O(e^{\log n \log \log n})$ operations (hard

Encryption, decryption, and key Generation in RSA



RSA Key Setup

- each user generates a public/private key pair by:
- selecting two large primes at random - p, q
- computing their system modulus $N=p.q$
  - note $\varnothing(N)=(p-1)(q-1)$
- selecting at random the encryption key e
  - where $1<e<\varnothing(N)$, $\gcd(e,\varnothing(N))=1$
- solve following equation to find decryption key d
  - $e.d=1 \bmod \varnothing(N)$ and $0 \leq d \leq N$
- publish their public encryption key: KU={e,N}
- keep secret private decryption key: KR={d,p,q}

RSA

- to encrypt a message M the sender:
  - obtains **public key** of recipient $KU=\{e,N\}$
  - computes: $C=M^e \bmod N$, where $0 \leq M < N$
- to decrypt the ciphertext C the owner:
  - uses their private key $KR=\{d,p,q\}$
  - computes: $M=C^d \bmod N$
- note that the message M must be smaller than the modulus N (block if needed)

Why RSA Works

- Euler's Theorem:

$$a^{\varnothing(n)} \bmod N = 1$$

  - where $g.c.d(a,N)=1$

- RSA have:
  - $N=p.q$
  - $\varnothing(N)=(p-1)(q-1)$
  - carefully chosen e & d to be inverses $\bmod \varnothing(N)$
  - hence $e.d=1+k.\varnothing(N)$ for some k

- Hence :

$$C^d = (M^e)^d = M^{1+k.\varnothing(N)} = M^1.(M^{\varnothing(N)})^q = M^1.(1)^q = M^1 = M \bmod N$$

RSA Example

1. Select primes: $p=17$ & $q=11$
2. Compute $n = pq = 17 \times 11 = 187$
3. Compute $\phi(n) = (p-1)(q-1) = 16 \times 10 = 160$
4. Select e: $\gcd(e,160)=1$; choose $e=7$
5. Determine d: $de=1 \bmod 160$ and $d < 160$ Value is $d=23$ since $23 \times 7 = 161 = 10 \times 160 + 1$
6. Publish public key $KU=\{7,187\}$
7. Keep secret private key $KR=\{23,17,11\}$

RSA Example cont.

- sample RSA encryption/decryption is:
- given message $M = 88$ (nb. $88<187$)
- encryption:

  $C = 88^7 \bmod 187 = 11$
- decryption:

  $M = 11^{23} \bmod 187 = 88$

RSA Key pair

(including Algorithm identifier)

[2048 bit]



Private Key

```
3082 010a 0282 0101 00b1 d311 e079 5543 0708 4ccb 0542 00e2 0d83 463d e493 bab6 06d3
0d59 bd3e c1ce 4367 018a 21a8 efbc ccd0 a2cc b055 9653 8466 0500 da44 4980 d854 0aa5
2586 94ed 6356 ff70 6ca3 a119 d278 be68 2a44 5e2f cfcc 185e 47bc 3ab1 463d 1ef0 b92c
345f 8c7c 4c08 299d 4055 eb3c 7d83 deb5 f0f7 8a83 0ea1 4cb4 3aa5 b35f 5a22 97ec 199b
c105 68fd e6b7 a991 942c e478 4824 1a25 193a eb95 9c39 0a8a cf42 b2f0 1cd5 5ffb 6bed
6856 7b39 2c72 38b0 ee93 a9d3 7b77 3ceb 7103 a938 4a16 6c89 2aca da33 1379 c255 8ced
9cbb f2cb 5b10 f82e 6135 c629 4c2a d02a 63d1 6559 b4f8 cdf9 f400 84b6 5742 859d 32a8
f92a 54fb ff78 41bc bd71 28f4 bb90 bcff 9634 04e3 459e a146 2840 8102 0301 0001
```

Public Key

```
3082 01e4 f267 0142 0f61 dd12 e089 5547 0f08 4ccb 0542 00e2 0d83 463d e493 bab6 0673
0d59 bf3e c1ce 4367 012a 11a8 efbc ccd0 a2cc b055 9653 8466 0500 da44 4980 d8b4 0aa5
2586 94ed 6356 ff70 6ca3 a119 d278 be68 2a44 5e2f cfcc 185e 47bc 3ab1 463d 1df0 b92c
345f 8c7c 4c08 299d 4055 eb3c 7d83 deb5 f0f7 8a83 0ea1 4cb4 3aa5 b35f 5a22 97ec 199b
c105 68fd e6b7 a991 942c e478 4824 1a25 193a eb95 9c39 0a8a cf42 b250 1cd5 5ffb 6b
6856 7b39 2c72 38b0 ee93 a9d3 7b77 3ceb 7103 a938 4a16 6c89 2aca da33 1379 c255 8c
9cbb f2cb 5b10 f82e 6135 c629 4c2a d02a 63d1 6559 b4f8 cdf9 f400 84b6 5742 859d 32a
f92a 54fb ff78 41bc bd71 28f4 bb90 bcff 9634 04de 45de af46 2240 8410 02f1 0001
```



Public key cryptosystem

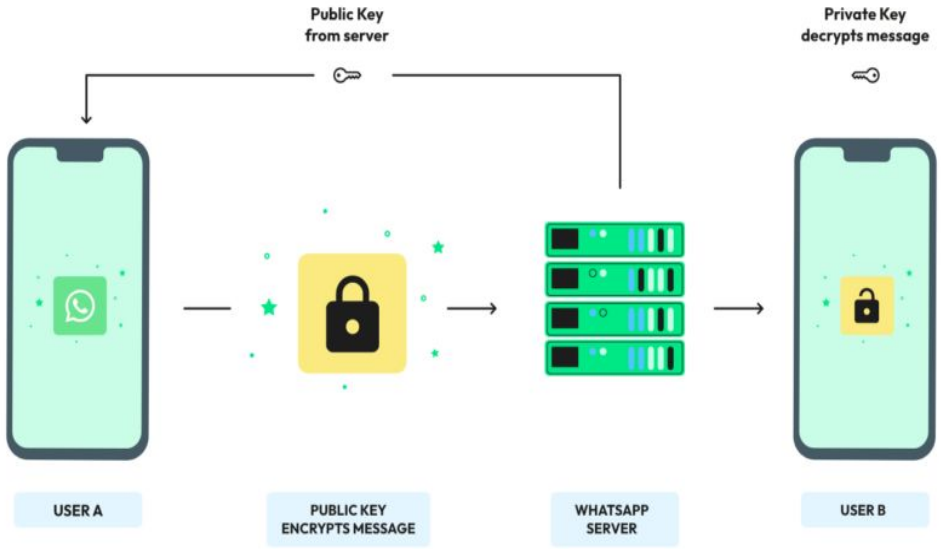
Previous Public key cryptography

1. RSA cryptosystem
2. ElGamal cryptosystem
3. Rabin cryptosystem
4. Diffie-Hellman Key exchange
5. Other (Lattice based cryptosystem, Elliptic curve-based cryptosystem)
6. Post-Quantum Cryptography

WhatsApp Security

- **Encryption Standard**: WhatsApp uses **Signal Protocol**, a cryptographic protocol developed by Open Systems Whispers.

- Signal Protocol is based on well-established cryptographic primitives and is widely regarded as one of the most secure messaging protocols available.



How WhatsApp encryptions works

Elliptic Curve Diffie-Hellman (ECDH):

- **ECDH** is used for **key exchange**.

AES (Advanced Encryption Standard):

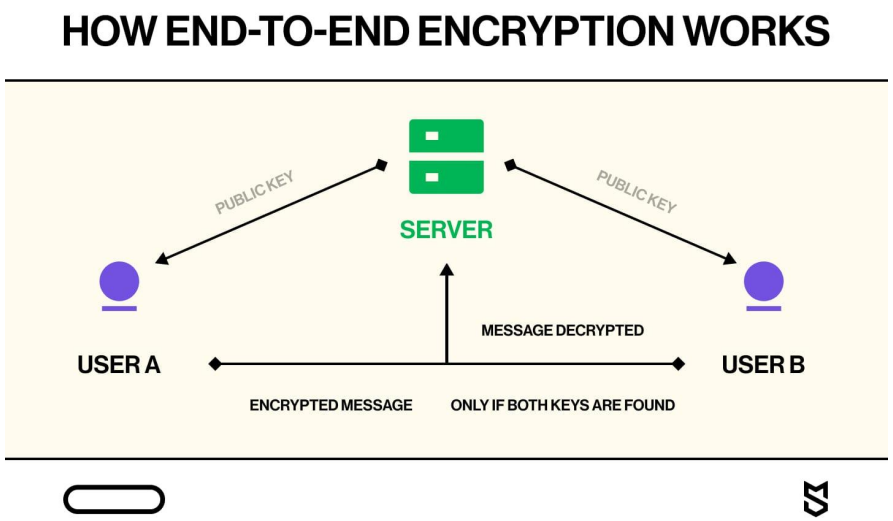
- WhatsApp uses **AES-256** in **Galois/Counter Mode (GCM)** for **symmetric encryption** of the message content.

HMAC (Hash-Based Message Authentication Code):

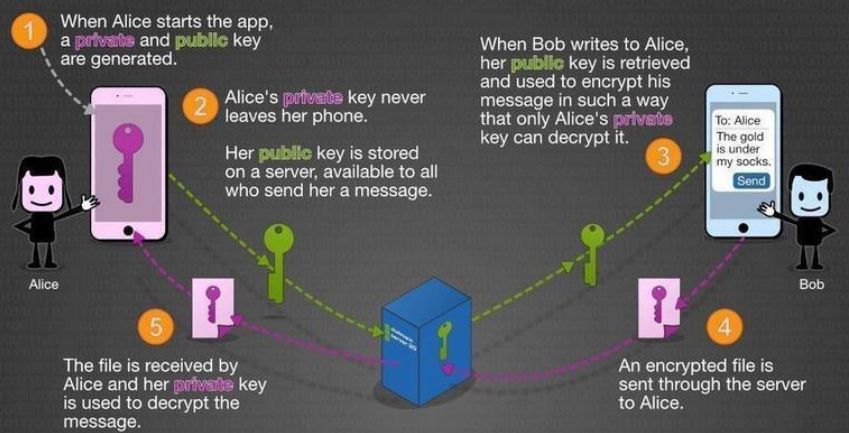
- **HMAC-SHA256** is used for **message integrity and authenticity**. It ensures that messages have not been tampered with during transmission.

EdDSA (Edwards-Curve Digital Signature Algorithm):

- **EdDSA** is used to sign messages and verify the authenticity of public keys.



End-to-End Encryption Explained



Prime Numbers & Encryption

$11 \times 17 = 187$

The product of 2 large random prime numbers is the backbone of encryption.

Cracking the encryption means figuring out the 2 factors. Using brute-force, it takes decades with today's computers. If the 2 numbers are known (a private key), a split second is all it takes.

17,425,170

The number of *digits* in the largest known prime number.

The public key is made up in part by calculating the number of integers that share no common factors, that are less than the product of the 2 prime numbers (encryption is supposed to be confusing).

Applications of End-to-End Encryption (E2EE) for Secure Communication

Instant messaging
Social Media Messaging
WhatsApp, Signal & iMessage

Email
Enhance privacy of users emails

VPNs
Protect data from hacking

Online Banking
Protect consumers' financial information privacy

Healthcare Institutions
Safeguard clinical information & sensible data of the patient.

Thank you